Depression and anxiety prevalence in people with cystic fibrosis and their caregivers: a systematic review and meta-analysis.

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Social Psychiatry and Psychiatric Epidemiology

Supplementary Data

Table S1. Embase and Medline database search strategy

1. exp depression/ and exp depressive disorder/
2. exp anxiety/
3. exp anxiety disorder/
4. (mood or psychology or psychiatry or emotion or mental health or psychological distress or nervous or hypervigilance).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word]
5. exp psychotropic agent/
6. 1 or 2 or 3 or 4 or 5
7. cystic fibrosis.mp. or exp cystic fibrosis/
8. 6 and 7

Table S2. CINAHL plus database search strategy

1. MH “Depression+”
2. MW depressive disorder
3. MH “Anxiety+” OR MH “Anxiety Disorders+”
4. Mood OR psychology OR psychiatry OR emotion OR mental health OR psychological distress
5. S1 OR S2 OR S3 OR S4
6. MH “Cystic Fibrosis”
7. Cystic fibrosis
8. S6 OR S7
9. S5 AND S8

Table S3. PsychINFO database search strategy

1. exp "Depression (Emotion)"/ or exp Major Depression/
2. Anxiety Disorders/ or exp Anxiety/
3. (mood or psychology or psychiat* or emotion or mental health or psychological distress or nervous or hypervigilance).mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures, mesh]
4. limit 3 to human
5. 1 or 2 or 4
6. cystic fibrosis.mp. or exp Cystic Fibrosis/
7. 5 and 6

Table S4. Study characteristics of included studies presented by participant group

Source	Publication Type	Location, income status	Total CF Participants (% male)	Mean age years (SD or range)	Age category used for analysis (years)	Depression tool, cut score	Anxiety tool, cut score
Children and Adolescents 0-18 years (36 studies)							
Allgood, 2017 [1]	Abstract	USA, HIC	7 (NR)	NR	12-18	CESD, 16	-
Blair, 2018 [2]	Abstract	USA, HIC	23 (60)	19.7 (0.6)	12-18	PHQ9, 5	GAD7, 5
Burke, 1989 [3]	Full publication	USA, HIC	52 (52)	12.19 (3.22)	Unclear	K-SADS, NR	-
Canter, 2017 [4]	Abstract	USA, HIC	45 (NR)	NR (all ≥12)	12-18	PHQ9, NR	GAD7, NR
Cronly, Duff, 2019 [5]	Full publication	Ireland, HIC	33 (NR)	NR (14-18)	12-18	HADS, 8	HADS, 8‡
Duff, 2018 [6]	Abstract	UK, HIC	125 (NR)	NR (≤16)	12-18	PHQ8, NR	GAD7, NR
Gravelle, 2018 [7]	Abstract	Canada, HIC	33 (70)	NR	12-18	PHQ9, NR	GAD7, NR
Graziano, 2017 [8]	Abstract	Italy, HIC	47 (36)	5.6 (1-11) children (N=67) 14.2 (12-17) adolescent (N=47)	12-18	PHQ9, 5	GAD7, 5
Graziano, 2020 [9]	Full publication	Italy, HIC	54 (37)	12-18	12-18	PHQ9, 5	GAD7, 5
Gundogdu, 2019 [10]	Full publication	Turkey, UMIC	32 (63)	8-16	5-11	CDI, 13	SCARED, 25
Harris, 2017 [11]	Abstract	USA, HIC	60 (depression); 28 (anxiety) (NR)	NR	12-18	PHQ9, NR	GAD7, NR
Hilliard, 2015 [12]	Full publication	USA, HIC	10 (NR)	16-18	12-18	CESD‡, 16	-
Hong, 2014 [13]	Abstract	USA, HIC	73 (44)	15.6 (NR)	12-18	RCADS, NR	-
Hosseinzadeh, 2019 [14]	Full publication	Iran, UMIC	40 (60)	5-18	5-11 12-18	K-SADS-PL, NR	††
Iturralde, 2017 [15]	Full publication	USA, HIC	27 (60)	12-22	12-18	PHQ9, 10	-
Laine, 2018 [16]	Abstract	Sweden, HIC	44 (50)	15 (10-18)	Unclear	BDI-Youth, NR	BAI-Youth, NR
Lee, 2017 [17]	Abstract	USA, HIC	81 (NR)	NR	12-18	PHQ9, NR	GAD7, NR
Liu, 2020 [18]	Full publication	USA, HIC	110 (NR)	12-17	12-18	PHQ9, 5	GAD7, 5
McLean, 2014 [19]	Abstract	USA, HIC	103 (41)	15.8 (2.9)	12-18	HADS, NR	HADS, NR
Modi, 2011 [20]	Full publication	USA, HIC	42 (NR)	12-18	12-18	HADS, 8‡	HADS, 8‡
Oliver, 2014 [21]	Full publication	USA, HIC	28 (NR)	NR (14-18)	12-18	HADS, 8‡	HADS, 8‡
Paida, 2020 [22]	Abstract	Australia, HIC	19 (NR)	8.2 (5)	Unclear	-	SCAS, NR
Pearson, 1991 [23]	Full publication	USA, HIC	61 (56)	10.96 (8-15)	Unclear	CDI, 13	STAI-Children, 1 Standard

							deviation above the mean
Pumariega, 1993 [24]	Full publication	USA, HIC	44 (60)	11 (7-15)	Unclear	CDI, 13	STAI-Children, 1 Standard deviation above the mean
Quittner, 2014 [25]	Full publication	Multiple	1286 (47)	14.84 (1.69)	12-18	HADS, 8; CESD§, 16	HADS, 8
Quittner, 2016 [26]	Abstract	USA, HIC	11 (NR)	14.9 (12-17)	12-18	PHQ9, 5	GAD7, 5
Smith, 2014 [27]	Full publication	USA, HIC	38 (47)	NR (7-17)	Unclear	CDI, 13	-
Smith, 2015 [28]	Abstract	USA, HIC	24 (NR)	NR	12-18	PHQ9, 5	-
Vandeleur, 2018 [29]	Full publication	Australia, HIC	75 (NR)	NR (7-18)	5-11 12-18	CDI, 55; BDI-Youth, 55	-
Verkleij, 2018 [30]	Full publication	Netherlands, HIC	17 (46)	NR (12-18)	12-18	PHQ9, 5	GAD7, 5
Walsh, 2017 [31]	Abstract	USA, HIC	83 (NR)	NR (12-18)	12-18	PHQ9, 5	GAD7, 5
Wray, 2004 [32]	Full publication	UK, HIC	Pre-transplant:12 (NR)†	NR (8-17)	12-18	MFQ-Children, 27; MFQ-Parent, 21	-
			Post-transplant: 11 (NR)	12-18	NA		
Adults (above 18 years) (54 studies)							
Afshar 2014 [33]	Abstract	USA, HIC	63 (65)	33.2 (NR)		PHQ9,5	-
Allgood, 2017 [1]	Abstract	USA, HIC	68 (NR)	NR		CESD, 16	-
Askew, 2017 [34]	Full publication	UK, HIC	45 (60)	20.7 (17-24)		HADS, 8	HADS, 8
Beenen, 2016 [35]	Abstract	USA, HIC	215 (NR)	NR		PHQ9, 5	GAD7, 5
Bowen, 2019 [36]	Abstract	USA, HIC	14 (NR)	36.2 (16-54)		PHQ9, 5	GAD7, 5
Bruschwein, 2018 [37]	Abstract	USA, HIC	19 (NR) (post admission results)	NR		-	GAD7, 5
Burge, 2015 [38]	Full publication	Australia, HIC	80 (100)	30 (8)		HADS, 8‡	HADS, 8‡
Catastini, 2018 [39]	Abstract	Italy, HIC	113 (57)	34.8 (9.7)		PHQ9, 5	GAD7, 5
Christon, 2017 [40]	Abstract	USA, HIC	95 (NR)	33 (18-73)		PHQ9, 5	GAD7, 5
Cronly, Duff, 2019 [5]	Full publication	Ireland, HIC	141 (NR)	32.2 (7.3) and 28.2 (10.6)		HADS, 8	HADS, 8‡
Delelis, 2008 [41]	Full publication	France, HIC	13 (NR)	28 (4.56)		CESD-French version, Male:17 Female 23	STAI-French version, Male:37 Female: 40
			3 with transplant (NR)				
Dvorak, 2013 [42]	Abstract	USA, HIC	100 (NR)	NR		CESD, 16	-

Fukushima, 2013 [43]	Abstract	USA, HIC	23 (70)	28.9 (19-50)		BDI, 17	-
Graziano, 2017 [8]	Abstract	Italy, HIC	90 (46)	27.4 (18-52)		PHQ9, 5	GAD7, 5
Graziano, 2020 [9]	Full publication	Italy, HIC	113 (44)	27 (8.2)		PHQ9, 5	GAD7, 5
Havermans, 2008 [44]	Full publication	Belgium, HIC	57 (51)	26.7 (8.1)		HADS, 8	HADS, 8
Havermans, 2011 [45]	Abstract	Belgium, HIC	35 (54)	NR		HADS, 8	HADS, 8
Hayee, 2019 [46]	Full publication	UK, HIC	107 (44)	28.8 (10.1) and 29.1 (8.8)		PHQ9, 10	GAD7, 10
Henderson, 2020 [47]	Full publication	Australia, HIC	39 (NR)	30 (10)		PHQ9, 5‡	GAD7, 5
Hilliard, 2015 [12]	Full publication	USA, HIC	128 (NR)	29.2 (16-63)		CESD‡, 16	-
Hjelm, 2017 [48]	Abstract	USA, HIC	189 (NR)	NR		PHQ9, NR	GAD7, NR
Iliza, 2014 [49]	Abstract	Canada, HIC	205 (49)	31.9 (9.2)		HADS, 8	HADS, 8
Knudsen, 2016 [50]	Full publication	Denmark, HIC	67 (41)	24.1 (18-30)		MDI, 20	-
Kopp, 2012 [51]	Abstract	USA, HIC	20 (NR)	26.9		CESD, 16	-
Kopp, 2013 [52]	Full publication	USA, HIC	30 (63)	26.1 (7)		CESD, 16	-
Kopp, 2016 [53]	Full publication	USA, HIC	30 (57)	27.7 (8.5)		QIDS-Self Report, 6; QIDS-Clinician, 6	-
Lalic, 2018 [54]	Abstract	Croatia, HIC	22 (NR) 6 with transplant (NR)	NR		PHQ9, 5	GAD7, 5
Lambrecht, 2011 [55]	Abstract	Belgium, HIC	324 (NR)	NR		HADS, 8	HADS, 8
Latchford, 2013 [56]	Full publication	UK, HIC	232 (57)	NR		HADS, 8; PHQ9, 5	-
Lee, 2017 [17]	Abstract	USA, HIC	105 (NR)	NR		PHQ9, NR	GAD7, NR
Modi, 2011 [20]	Full publication	USA, HIC	17 (NR)	NR		HADS, 8‡	HADS, 8‡
Oliver, 2014 [21]	Full publication	USA, HIC	44 (NR)	19.1 (3.3)		HADS, 8‡	HADS, 8‡
Olveira, Giron 2010 [57]	Abstract	Spain, HIC	79 (47)	25.9 (7.7)		HADS, 8 CESD, 16	HADS, 8
Olveira, Olveira 2010 [58]	Abstract	Spain, HIC	43 (NR)	NR		HADS, 8 CESD, 16	HADS, 8
Orava, 2018 [59]	Full publication	Canada, HIC	22 (46)	33 (18-67)		HADS, 8‡	-
Pakhale, 2015 [60]	Full publication	Canada, HIC	45 (58)	30.7 (10.8)		CESD, 16	GAD7, 10
Pearson, 1991 [23]	Full publication	USA, HIC	36 (56)	24.81 (16-40)		Zung, 50	STAI, 1 Standard deviation above the mean
Quittner, 2014 [25]	Full publication	Multiple	4739 (52)	28.87 (9.5)		HADS, 8; CESD§, 16	HADS, 8

Quittner, 2016 [26]	Abstract	USA, HIC	56 (NR)	31.7 (18-72)		PHQ9, 5	GAD7, 5
Quon, 2015 [61]	Full publication	USA, HIC	153 (50)	28.6 (8.8)		PHQ9, 10	GAD7, 10
Rached, 2011 [62]	Abstract	Brazil, UMIC	16 (56)	27.9		HADS, NR	HADS, NR
Riekert, 2007 [63]	Full publication	USA, HIC	76 (45)	NR		BDI, 10	-
Rightmer, 2017 [64]	Abstract	USA, HIC	65 (NR)	NR		PHQ9, 5	GAD7, 5
Sandage, 2015 [65]	Abstract	USA, HIC	59 (37)	25		PHQ9, NR	GAD7, NR
Sherman, 2020 [66]	Full publication	USA, HIC	69 (57)	27.51 (9.62)		HADS, 8	-
Smith, 2015 [28]	Abstract	USA, HIC	89 (NR)	NR		PHQ9, 5	-
Talbot, 2012 [67]	Abstract	UK, HIC	326 (48)	30.9 (10.26)		HADS, NR	HADS, NR
Uslu, 2018 [68]	Abstract	Turkey, UMIC	30 (43)	24 (4)		HADS, 8	HADS, 10
Walker, 2015 [69]	Abstract	USA, HIC	74 (47)	NR		PDSF, NR	PASF, NR
Weldon, 2019 [70]	Abstract	UK, HIC	101 (NR)	NR		PHQ9, NR	GAD7, NR
Westell, 2014 [71]	Abstract	Canada, HIC	58 (NR)	NR		CESD, 16	GAD7, NR
Wolfe, 2016 [72]	Abstract	USA, HIC	60 (NR)	NR		PHQ9, NR	GAD7, NR
Wood, 2020 [73]	Full publication	Australia, HIC	60 (41)	31 (9)		HADS, 8‡	HADS, 8‡
Yohannes, 2012 [74]	Full publication	UK, HIC	121 (54)	30 (18-70)		HADS, 8‡	HADS, 8‡
Caregivers (30 studies)							
Akca, 2016 [75]	Abstract	Turkey, UMIC	40 (0)	NR		BDI, 10	STAI, NR
Barker, 2016 [76]	Full publication	USA, HIC	83 (NR)	NR		CESD, 16	-
Beinke, 2016 [77]	Full publication	Australia, HIC	51 (0)	48.83 (4.34)		DASS, 10	DASS, 10
Betz, 2019 [78]	Full publication	USA, HIC	23 (22)	NR		CESD, 16	-
Bhat, 2018 [79]	Full publication	India, LMIC	35 (66)	Mothers 27.8 (4.7) Fathers 30.9 (5.4)		CESD, 16	-
Branch-Smith, 2018 [80]	Abstract	Australia, HIC	47 (NR)	NR		DASS, NR	DASS, NR
Casana-Granell, 2018 [81]	Full publication	Spain, HIC	13 (NR)	NR		HADS, 8‡	HADS, 8‡
Cronly, Horgan, 2019 [82]	Full publication	Ireland, HIC	203 (25)	29.33 (6.6)		HADS, 8; CESD, 16	HADS, 8
Driscoll, 2010 [83]	Full publication	USA, HIC	87 (13)	36.61 (8.4)		CESD, 16	-
Duff, 2018 [6]	Abstract	UK, HIC	143 (NR)	NR		PHQ8, NR	GAD7, NR
Glasscoe, 2007 [84]	Full publication	UK, HIC	80 (59)#	NR		BDI, 13	-
Goetz, 2016 [85]	Abstract	USA, HIC	41 (NR)	NR		PHQ9, NR	-
Goodfellow, 2015 [86]	Full publication	UK, HIC	90 (NR)	NR		CESD, 16	-
Gravelle, 2018 [7]	Abstract	Canada, HIC	20 (30)	NR		PHQ9, NR	GAD7, NR
Graziano, 2017 [8]	Abstract	Italy, HIC	168 (38)	Mothers 40.1 (22-58); Fathers 41.1 (24-53)		PHQ9, 5	GAD7, 5

Graziano, 2020 [9]	Full publication	Italy, HIC	186 (37)	Mothers 41 (7.5) Fathers 41 (7.1)		PHQ9, 5	GAD7, 5
Grossoehme, 2015 [87]	Full publication	USA, HIC	142 (28)	NR		CESD, 16	-
Lambrecht, 2011 [55]	Abstract	Belgium, HIC	348 (31)	NR		HADS, 8	HADS, 8
Lee, 2017 [17]	Abstract	USA, HIC	196 (depression); 192 (anxiety) (NR)	NR		PHQ9, NR	GAD7, NR
Modi, 2011 [20]	Full publication	USA, HIC	40 (10)	43.18 (5.5)		HADS, 8‡	HADS, 8‡
Naranjo, 2017 [88]	Abstract	USA, HIC	102 (NR)	NR		PHQ9, 10	GAD7, 10
Quittner, 2014 [25]	Full publication	Multiple	4102 (24)	NR		HADS, 8; CESD§, 16	HADS, 8
Quittner, 2016 [26]	Abstract	USA, HIC	9 (NR)	NR		PHQ9, 5	GAD7, 5
Sheehan, 2012 [89]	Full publication	Australia, HIC	102 (54)	37.5 (26.6 - 61.6)		DASS, 10	DASS, 8
Szentpetery, 2018 [90]	Abstract	USA, HIC	54 (NR)	37.5 (9.1)		PHQ8, 5	GAD7, 5
Tluczek, 2014 [91]	Full publication	USA, HIC	33 (6)	48.1 (4.7) and 49.2 (7.9)		CESD, 16	-
Unal, 2020 [92]	Full publication	Turkey, UMIC	36 (0)	35.1 (1.2)		BDI, 10‡	-
Verkleij, 2018 [30]	Full publication	Netherlands, HIC	79 (42)	Mothers 40 (6.6); Fathers 42 (7.5)		PHQ9, 5	GAD7, 5
Walsh, 2018 [93]	Abstract	USA, HIC	29 (NR)	NR		PHQ9, 5	GAD7, 5
Yilmaz, 2008 [94]	Full publication	Turkey, UMIC	21 (0)	NR		HADS, 8‡	HADS, 8‡

Abbreviations; BAI: Beck Anxiety Inventory; BDI: Beck Depression Inventory; CESD: Center for Epidemiologic Studies Depression Scale; CDI: Children's Depression Inventory; DASS: Depression Anxiety Stress Scale; GAD7: Generalized Anxiety Disorder 7-item measure; HADS: The Hospital Anxiety and Depression Scale; K-SADS: Kiddie Schedule for Affective Disorders and Schizophrenia; K-SADS-PL: Kiddie Schedule for Affective Disorders and Schizophrenia-Present and Lifetime version; MDI: Major Depression Inventory; MFQ: Mood and Feelings Questionnaire; NR: not reported, unclear or not reported specifically for the CF subset used in this meta-analysis; PASF: PROMIS Anxiety Short Form. PDSF: PROMIS Depression Short Form; PHQ: Patient Health Questionnaire; QIDS: Quick Inventory of Depressive Symptomatology; RCADS: Revised Children's Anxiety and Depression Survey; SCARED: Screen for Child Anxiety Related Disorders; SCAS: The Spence Children's Anxiety Scale; STAI: State-Trait Anxiety Inventory; Unclear refers to studies where participant age is not exclusive to either 5-11 or 12-18 years of age.

†Data analysed from pre-transplantation participant sample. ‡ Data obtained from personal correspondence with author. §Data was not collected on both tools by all participants. ¶Cut score not comparable to other studies in this analysis. #Data analysed from 9-month sample. ††Multiple anxiety diagnoses reported, unable to differentiate between acute and chronic states, therefore not used in meta-analysis.

Risk of bias critical appraisal assessment

For sample size adequacy, a minimum number of three or seven participants (for studies involving participants <18 or ≥18 years of age, respectively) was set by reviewers prior to appraisal. Numbers were determined using a minimum sample size calculation based on expected prevalence values, conservatively taken from the findings of the largest study to date in this subject matter. [95,25] Unless an article scored less than 50% of the total score possible, it was deemed not at high risk of bias and was included for systematic review and meta-analysis. Note that for each question deemed not applicable for a study, the denominator for the overall score was lessened by 1, thereby not penalising the study for a lower score achieved.

Table S5. Risk of bias assessment for analytical cross-sectional studies

For sample size adequacy, a minimum number of three or seven participants (for studies involving participants <18 or ≥18 years of age, respectively) was set by reviewers prior to appraisal. Numbers were determined using a minimum sample size calculation based on expected prevalence values, conservatively taken from the findings of the largest study to date in this subject matter. [95,25] Unless an article scored less than 50% of the total score possible, it was deemed not at high risk of bias and was included for systematic review and meta-analysis.

Citation	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Total / 8
Barker, 2016.	Y	Y	Y	Y	N	N	Y	Y	6
Beinke, 2017.	U	Y	Y	Y	Y	U	Y	Y	6
Bhat, 2018.	Y	N	Y	Y	Y	N	Y	Y	6
Branch-Smith, 2018.	U	N	Y	Y	N	N/A	Y	Y	4
Burge, 2015.	Y	Y	Y	Y	U	U	U	Y	4
Cronly, Duff, 2019.	Y	Y	Y	Y	Y	U	Y	Y	7
Delelis, 2008.	Y	Y	Y	Y	Y	Y	Y	Y	8
Goodfellow, 2015.	Y	Y	Y	Y	Y	Y	Y	Y	8
Grossoehme, 2015.	N	U	Y	Y	Y	N	U	Y	4
Havermans, 2008.	N	N	U	Y	Y	Y	Y	U	4
Hayee, 2019.	Y	U	Y	Y	N	U	Y	Y	5
Henderson, 2020.	U	Y	Y	Y	N	N	Y	Y	5
Hilliard, 2015.	Y	Y	Y	Y	U	U	Y	Y	6
Iturralde, 2017.	Y	U	Y	Y	Y	N	Y	Y	6
Liu, 2020.	Y	Y	Y	Y	U	U	Y	Y	6
Orava, 2018.	Y	Y	Y	Y	Y	Y	Y	Y	8
Pakhale, 2015.	Y	Y	Y	Y	N	N	Y	Y	6
Pumariega, 1993.	Y	N	Y	Y	N	N	Y	U	4
Riekert, 2007.	U	Y	Y	Y	N	N	Y	Y	5
Sherman, 2020.	Y	U	Y	Y	U	N/A	Y	Y	5
Smith, 2014.	Y	N	Y	Y	Y	U	Y	Y	6
Tluczek, 2014.	Y	Y	Y	Y	Y	U	Y	Y	7
Verkleij, 2018.	Y	Y	Y	Y	U	U	Y	Y	6
Yilmaz, 2008.	Y	N	Y	Y	U	U	U	Y	4
%	71	58	96	100	46	17	88	92	

1. Were the criteria for inclusion in the sample clearly defined? 2. Were the study subjects and the setting described in detail? 3. Was the exposure measured in a valid and reliable way? 4. Were objective, standard criteria used for measurement of the condition? 5. Were confounding factors identified? 6. Were strategies to deal with confounding factors stated? 7. Were the outcomes measured in a valid and reliable way? 8. Was appropriate statistical analysis used?

Table S6. Risk of bias assessment for prevalence studies

For sample size adequacy, a minimum number of three or seven participants (for studies involving participants <18 or ≥18 years of age, respectively) was set by reviewers prior to appraisal. Numbers were determined using a minimum sample size calculation based on expected prevalence values, conservatively taken from the findings of the largest study to date in this subject matter. [95,25] Unless an article scored less than 50% of the total score possible, it was deemed not at high risk of bias and was included for systematic review and meta-analysis.

Citation	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Total / 9
Askew, 2017.	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Burke, 1989.	Y	U	Y	N	U	Y	Y	Y	U	5
Cronly, Horgan, 2019	Y	Y	Y	Y	Y	Y	Y	Y	N	8
Driscoll, 2010.	Y	Y	Y	Y	U	Y	U	Y	U	6
Graziano, 2020.	Y	Y	Y	Y	Y	Y	Y	U	Y	7
Gundogdu, 2019.	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Hosseinzadeh, 2019.	Y	N	Y	N	Y	Y	U	U	Y	5
Knudsen, 2016.	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Kopp, 2013.	Y	U	Y	Y	N	Y	Y	Y	U	6
Latchford, 2013.	Y	Y	Y	N	U	Y	Y	U	U	5
Modi, 2011.	Y	N	Y	Y	Y	Y	U	Y	Y	7
Oliver, 2014.	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Pearson, 1991.	Y	U	Y	U	Y	Y	N	Y	U	5
Quittner, 2014.	Y	Y	Y	Y	Y	Y	N	Y	Y	8
Quon, 2015.	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Unal, 2020.	Y	Y	Y	U	Y	Y	Y	Y	N	7
Vandeleur, 2018.	Y	U	Y	U	Y	Y	U	Y	U	5
Wray, 2004.	Y	Y	Y	U	Y	Y	Y	Y	Y	8
Yohannes, 2012.	Y	U	Y	Y	Y	Y	Y	Y	Y	8
%	100	63	95	63	79	100	68	84	58	

1. Was the sample frame appropriate to address the target population? 2. Were study participants sampled in an appropriate way? 3. Was the sample size adequate? 4. Were the study subjects and the setting described in detail? 5. Was the data analysis conducted with sufficient coverage of the identified sample? 6. Were valid methods used for the identification of the condition? 7. Was the condition measured in a standard, reliable way for all participants? 8. Was there appropriate statistical analysis? 9. Was the response rate adequate, and if not, was the low response rate managed appropriately?

Table S7. Risk of bias assessment for diagnostic test accuracy studies

For sample size adequacy, a minimum number of three or seven participants (for studies involving participants <18 or ≥18 years of age, respectively) was set by reviewers prior to appraisal. Numbers were determined using a minimum sample size calculation based on expected prevalence values, conservatively taken from the findings of the largest study to date in this subject matter. [95,25] Unless an article scored less than 50% of the total score possible, it was deemed not at high risk of bias and was included for systematic review and meta-analysis.

Citation	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Total / 10
Casana-Granell, 2018.	Y	Y	Y	Y	N/A	N/A	N/A	U	Y	Y	6

1. Was a consecutive or random sample of patients enrolled? 2. Was a case control design avoided? 3. Did the study avoid inappropriate exclusions? 4. Were the index test results interpreted without knowledge of the results of the reference standard? 5. If a threshold was used, was it pre-specified? 6. Is the reference standard likely to correctly classify the target condition? 7. Were the reference standard results interpreted without knowledge of the results of the index test? 8. Was there an appropriate interval between index test and reference standard? 9. Did all patients receive the same reference standard? 10. Were all patients included in the analysis?

Table S8. Risk of bias assessment for cohort studies

For sample size adequacy, a minimum number of three or seven participants (for studies involving participants <18 or ≥18 years of age, respectively) was set by reviewers prior to appraisal. Numbers were determined using a minimum sample size calculation based on expected prevalence values, conservatively taken from the findings of the largest study to date in this subject matter. [95,25] Unless an article scored less than 50% of the total score possible, it was deemed not at high risk of bias and was included for systematic review and meta-analysis.

Citation	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Total / 11
Glasscoe, 2007.	Y	Y	Y	Y	Y	U	Y	Y	N	N	Y	8
Kopp, 2016.	Y	Y	Y	Y	Y	U	Y	Y	Y	N/A	Y	9
Sheehan, 2012.	N	Y	Y	U	U	U	Y	Y	Y	N	Y	6
%	67	100	100	67	67	0	100	100	67	0	100	

1. Were the two groups similar and recruited from the same population? 2. Were the exposures measured similarly to assign people to both exposed and unexposed groups? 3. Was the exposure measured in a valid and reliable way? 4. Were confounding factors identified? 5. Were strategies to deal with confounding factors stated? 6. Were the groups/participants free of the outcome at the start of the study (or at the moment of exposure)? 7. Were the outcomes measured in a valid and reliable way? 8. Was the follow up time reported and sufficient to be long enough for outcomes to occur? 9. Was follow up complete, and if not, were the reasons to loss to follow up described and explored? 10. Were strategies to address incomplete follow up utilized? 11. Was appropriate statistical analysis used?

Table S9. Risk of bias assessment for randomized controlled trial studies

For sample size adequacy, a minimum number of three or seven participants (for studies involving participants <18 or ≥18 years of age, respectively) was set by reviewers prior to appraisal. Numbers were determined using a minimum sample size calculation based on expected prevalence values, conservatively taken from the findings of the largest study to date in this subject matter. [95,25] Unless an article scored less than 50% of the total score possible, it was deemed not at high risk of bias and was included for systematic review and meta-analysis.

Citation	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Total / 13
Betz, 2019.	Y	N	Y	N	N	N	Y	N	Y	Y	Y	Y	U	7
Wood, 2020.	Y	Y	U	N	Y	U	Y	Y	Y	Y	Y	Y	Y	10
%	100	50	0	0	50	0	100	50	100	100	100	100	50	

1. Was true randomization used for assignment of participants to treatment groups? 2. Was allocation to treatment groups concealed? 3. Were treatment groups similar at the baseline? 4. Were participants blind to treatment assignment? 5. Were those delivering treatment blind to treatment assignment? 6. Were outcomes assessors blind to treatment assignment? 7. Were treatment groups treated identically other than the intervention of interest? 8. Was follow up complete and if not, were differences between groups in terms of their follow up adequately described and analysed? 9. Were participants analysed in the groups to which they were randomized? 10. Were outcomes measured in the same way for treatment groups? 11. Were outcomes measured in a reliable way? 12. Was appropriate statistical analysis used? 13. Was the trial design appropriate, and any deviations from the standard RCT design (individual randomization, parallel groups) accounted for in the conduct and analysis of the trial?

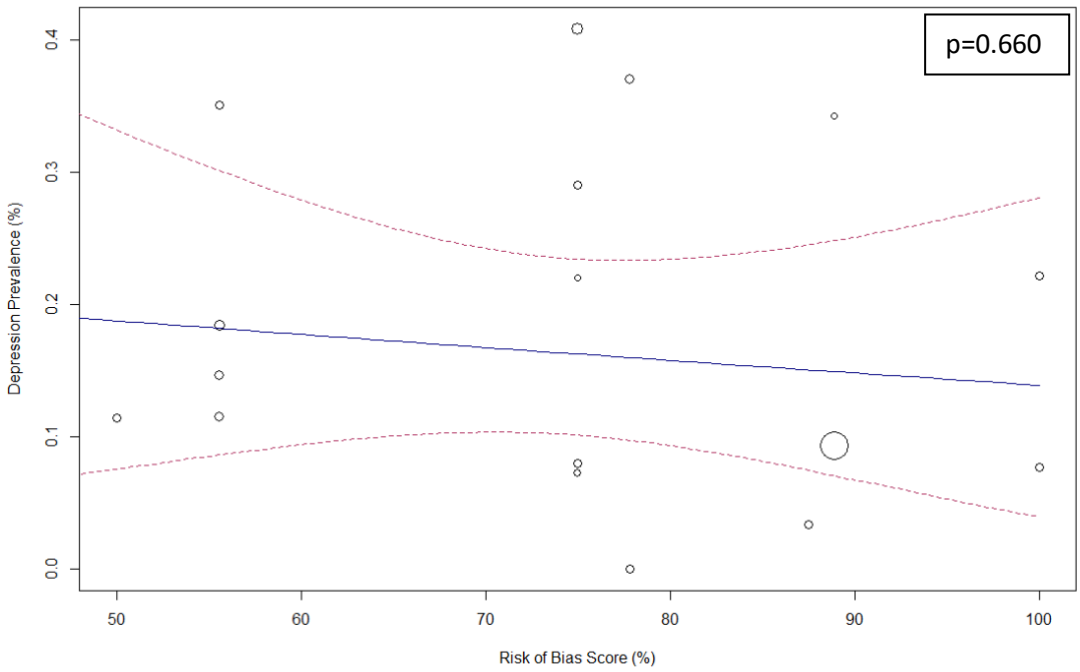

Figure S1. Meta-regression of risk of bias and depression prevalence in children and adolescents with cystic fibrosis

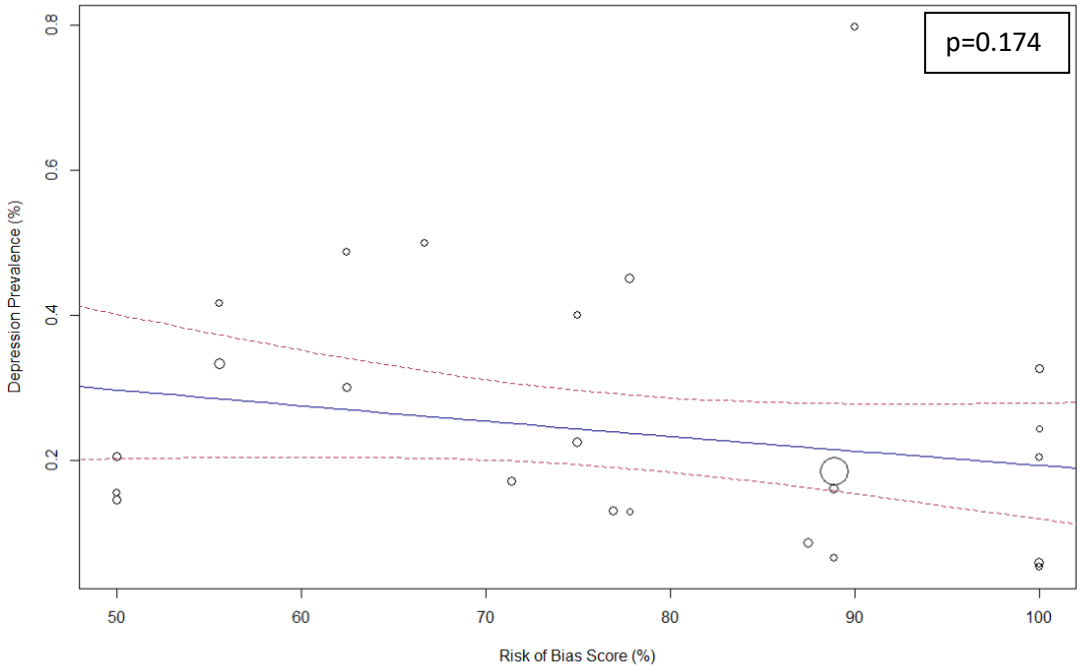

Figure S2. Meta-regression of risk of bias and depression prevalence in adults with cystic fibrosis

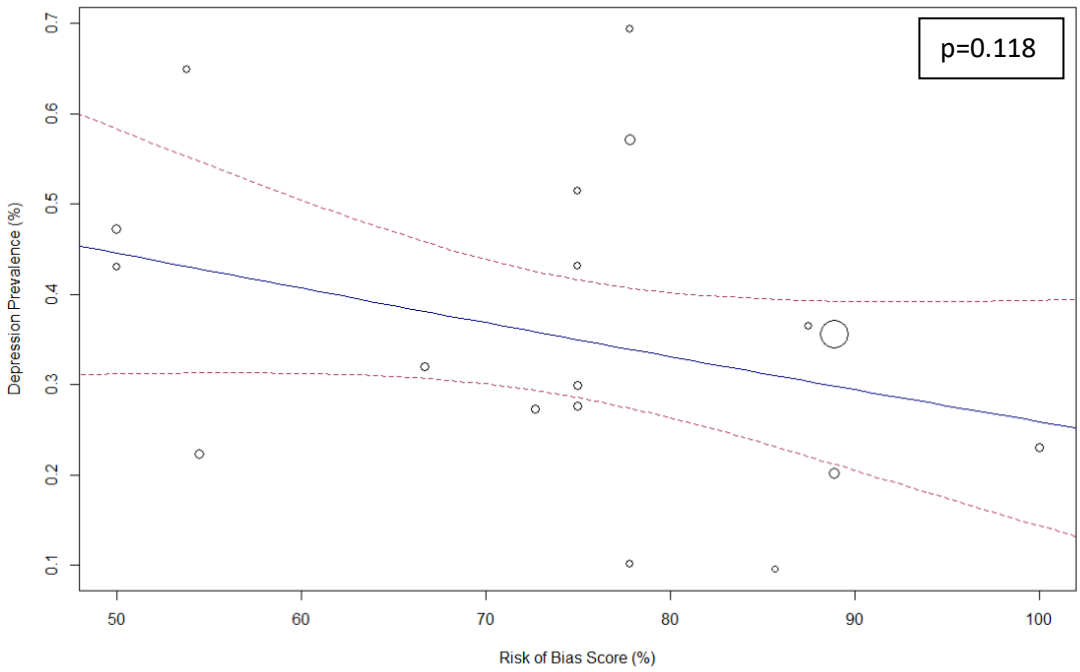

Figure S3. Meta-regression of risk of bias and depression prevalence in caregivers of people with cystic fibrosis

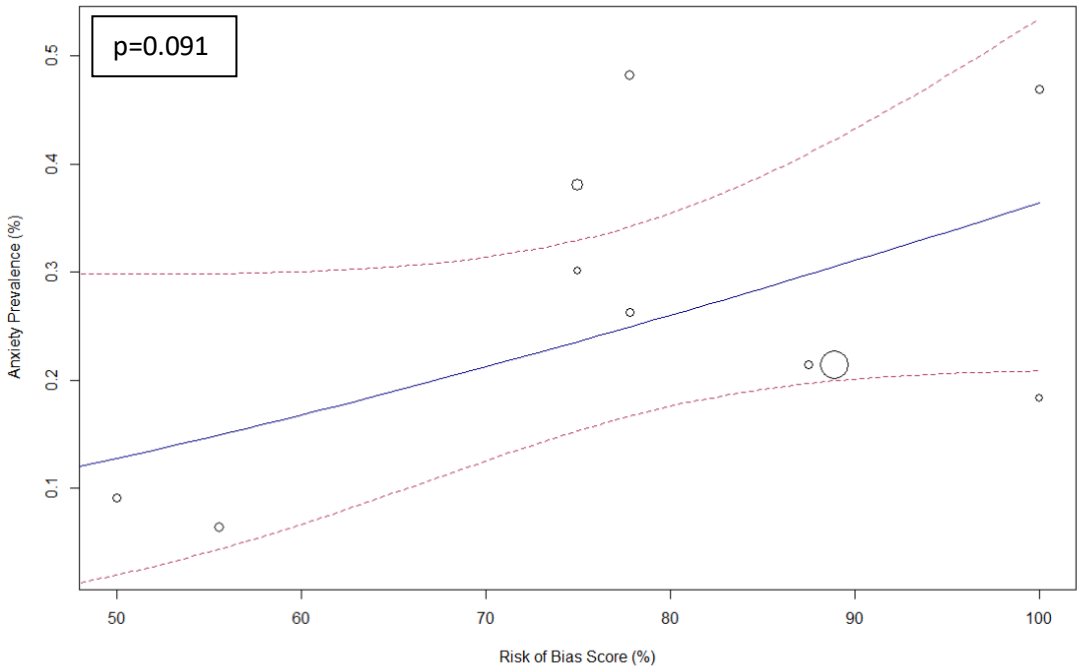

Figure S4. Meta-regression of risk of bias and anxiety prevalence in children and adolescents with cystic fibrosis

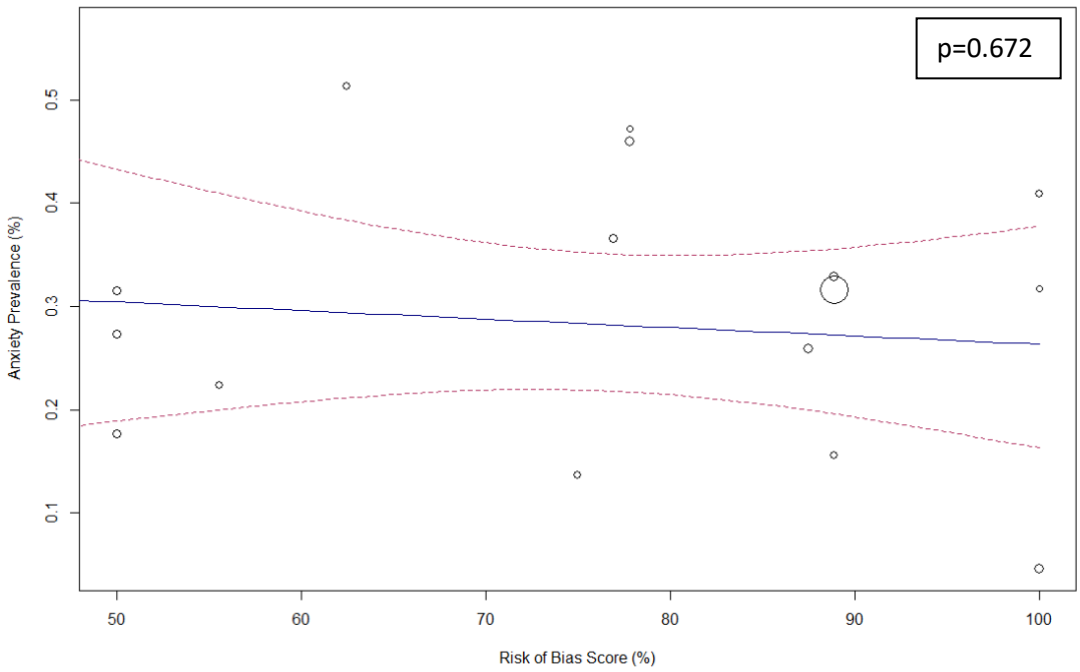

Figure S5. Meta-regression of risk of bias and anxiety prevalence in adults with cystic fibrosis

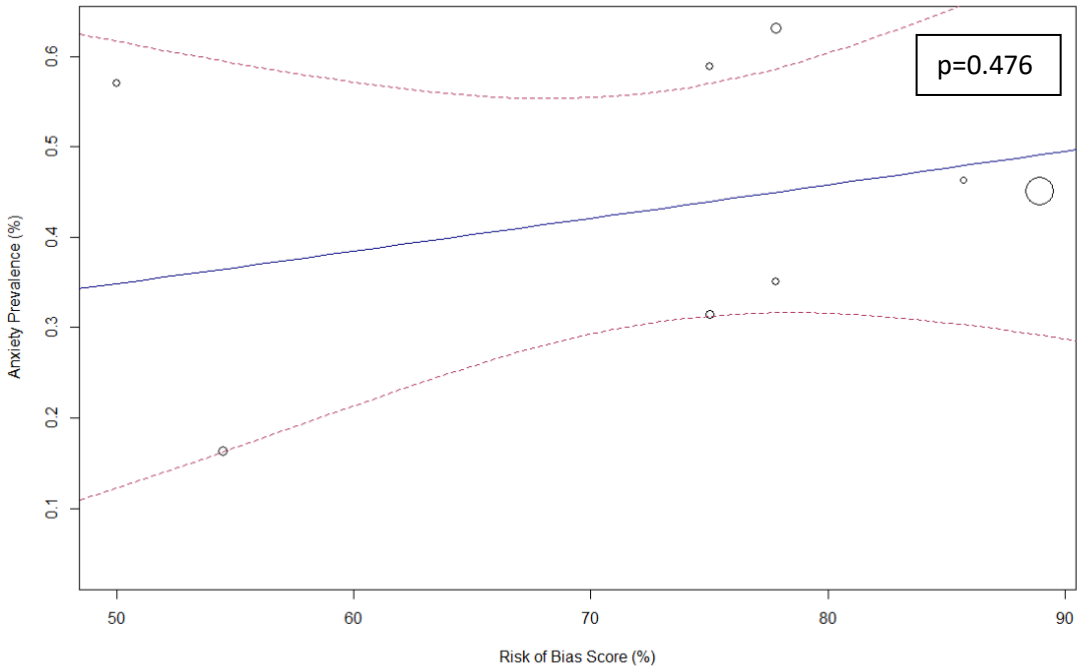

Figure S6. Meta-regression of risk of bias and anxiety prevalence in caregivers of people with cystic fibrosis

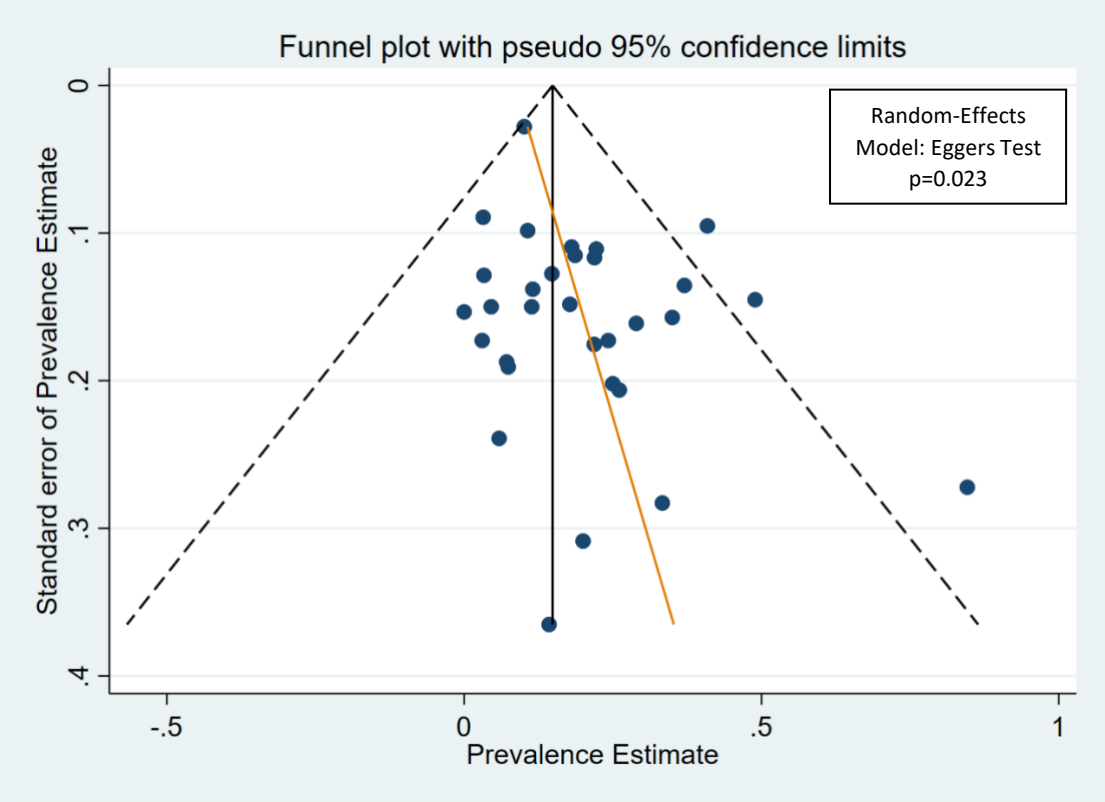

Figure S7. Depression prevalence in children and adolescents with cystic fibrosis

Prevalence estimate represented where 0.5=50%, 1=100%

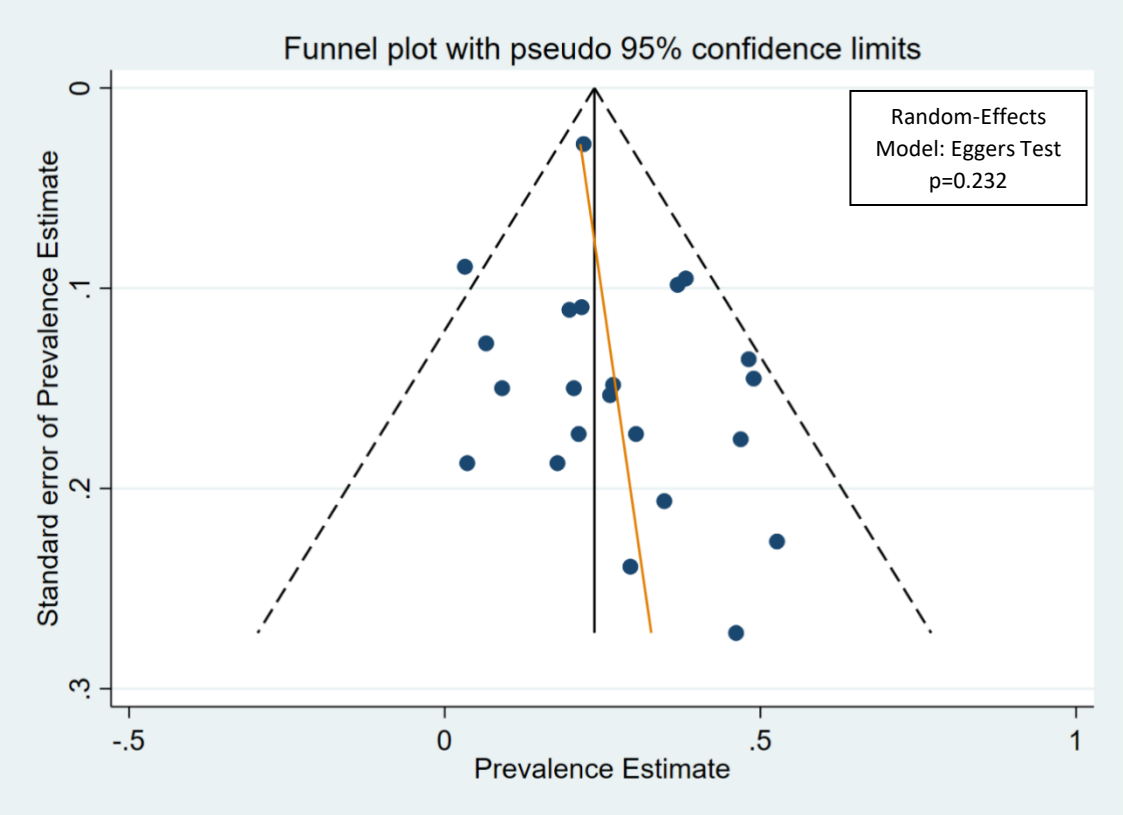

Figure S8. Anxiety prevalence in children and adolescents with cystic fibrosis

Prevalence estimate represented where 0.5=50%, 1=100%

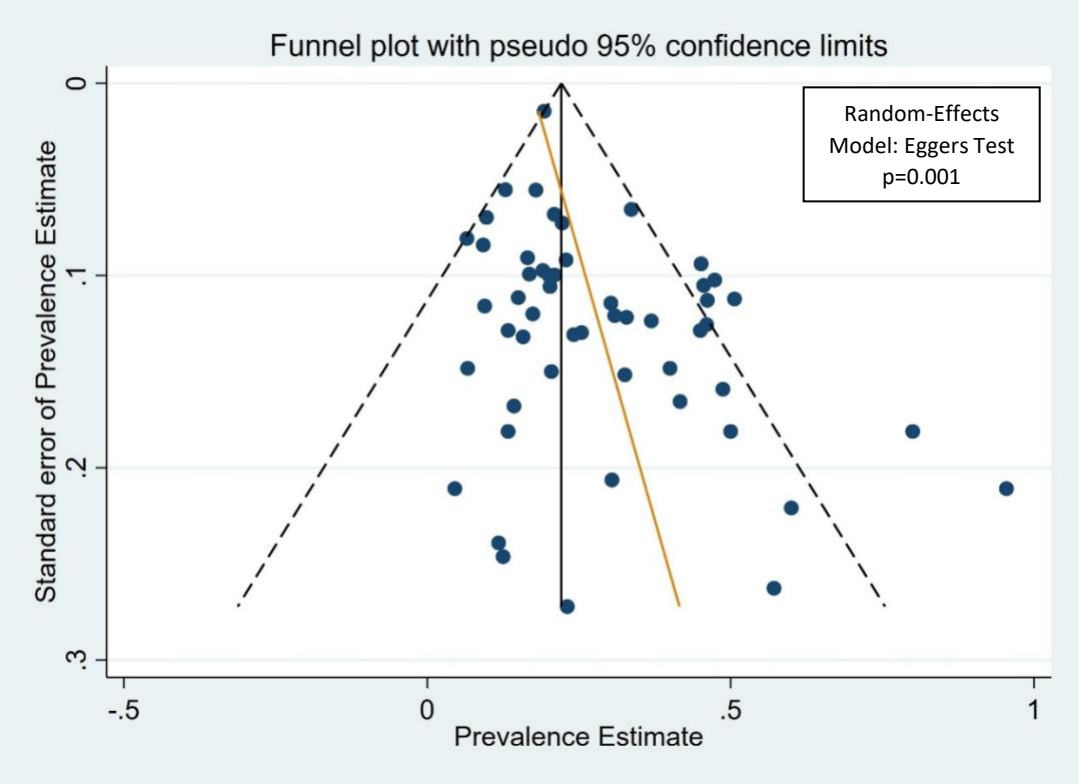

Figure S9. Depression prevalence in adults with cystic fibrosis

Prevalence estimate represented where 0.5=50%, 1=100%

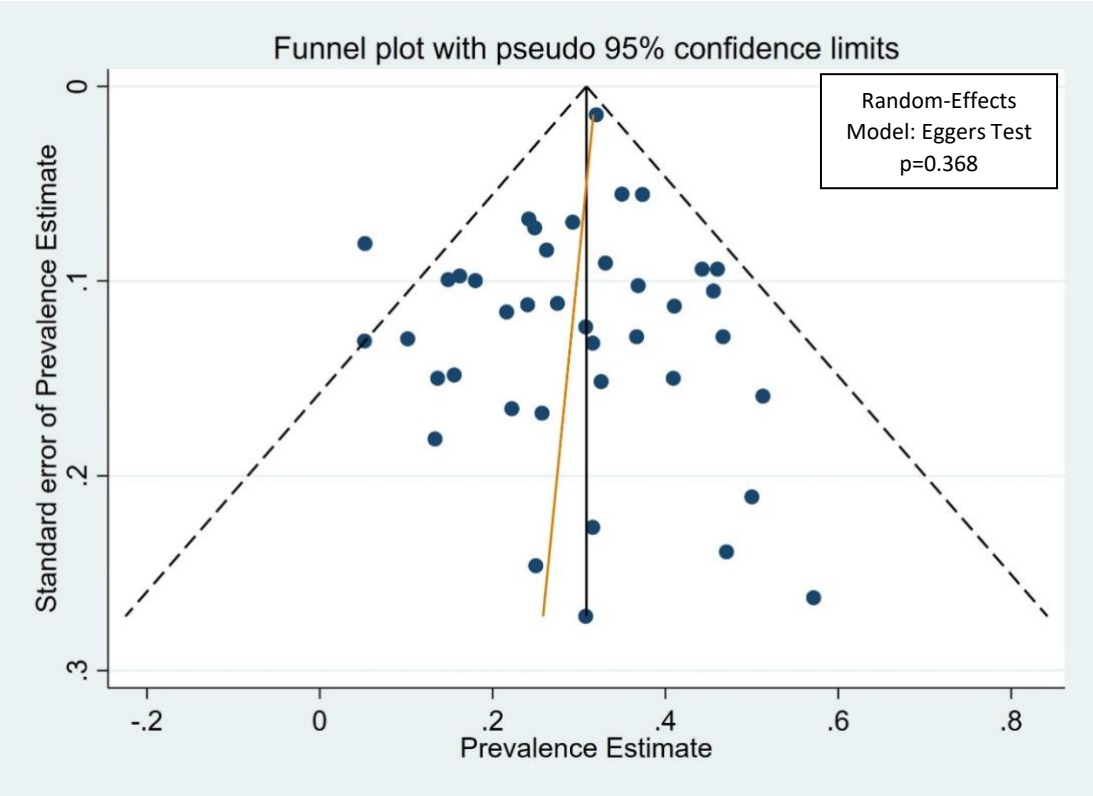

Figure S10. Anxiety prevalence in adults with cystic fibrosis

Prevalence estimate represented where 0.5=50%, 1=100%

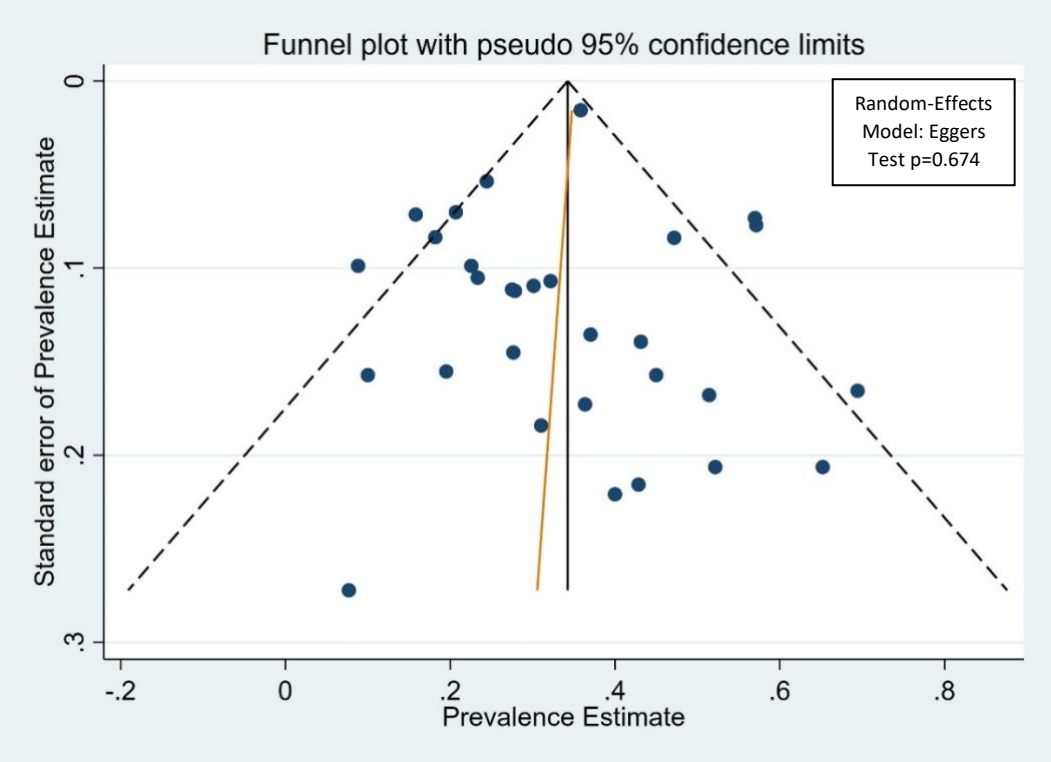

Figure S11. Depression prevalence in caregivers of people with cystic fibrosis

Prevalence estimate represented where 0.5=50%, 1=100%

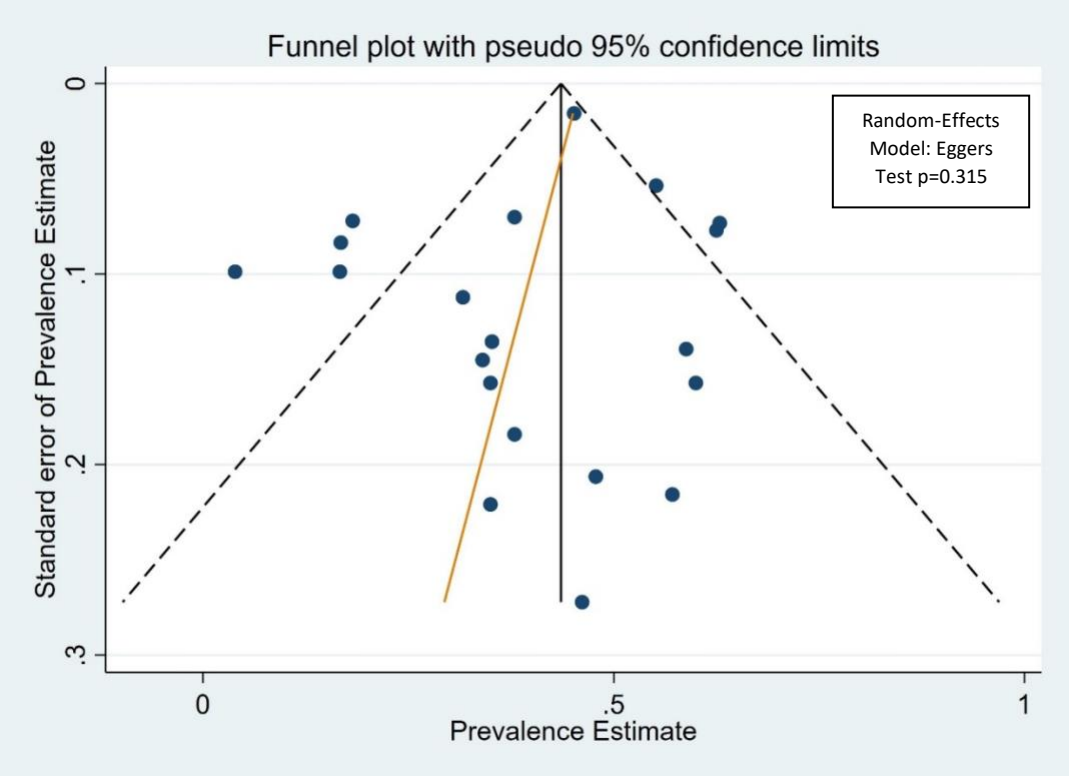

Figure S12. Anxiety prevalence in caregivers of people with cystic fibrosis

Prevalence estimate represented where 0.5=50%, 1=100%

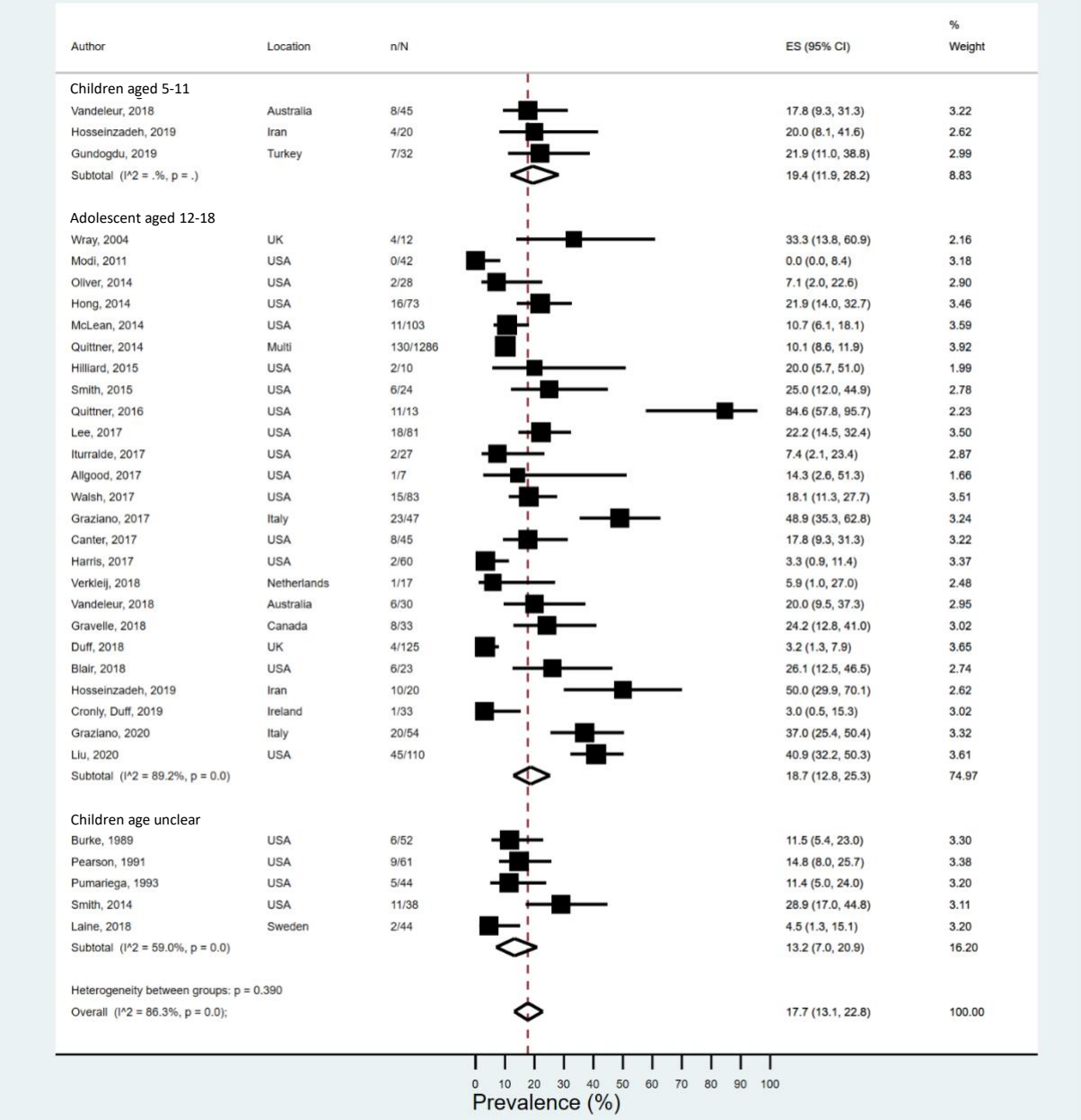

Figure S13. Forest plot of pooled prevalence of depression in children and adolescents with cystic fibrosis by age group

Abbreviations: CI: Confidence Interval; ES: Effect Size. Children age unclear refers to studies where participant age is not exclusive to either 5-11 or 12-18 years of age.

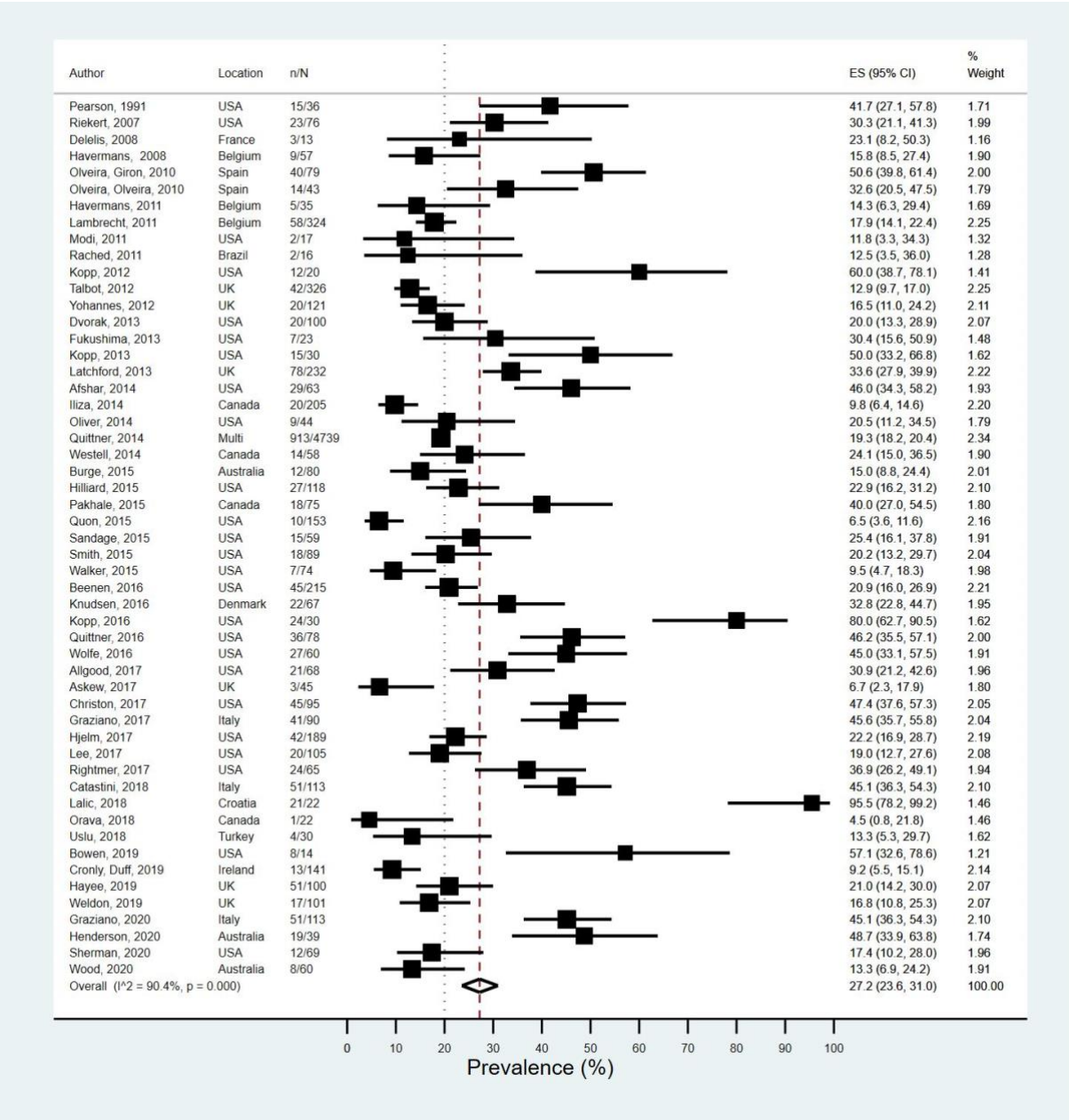

Figure S14. Forest plot of pooled prevalence of depression in adults with cystic fibrosis

Abbreviations: CI: Confidence Interval; ES: Effect Size.

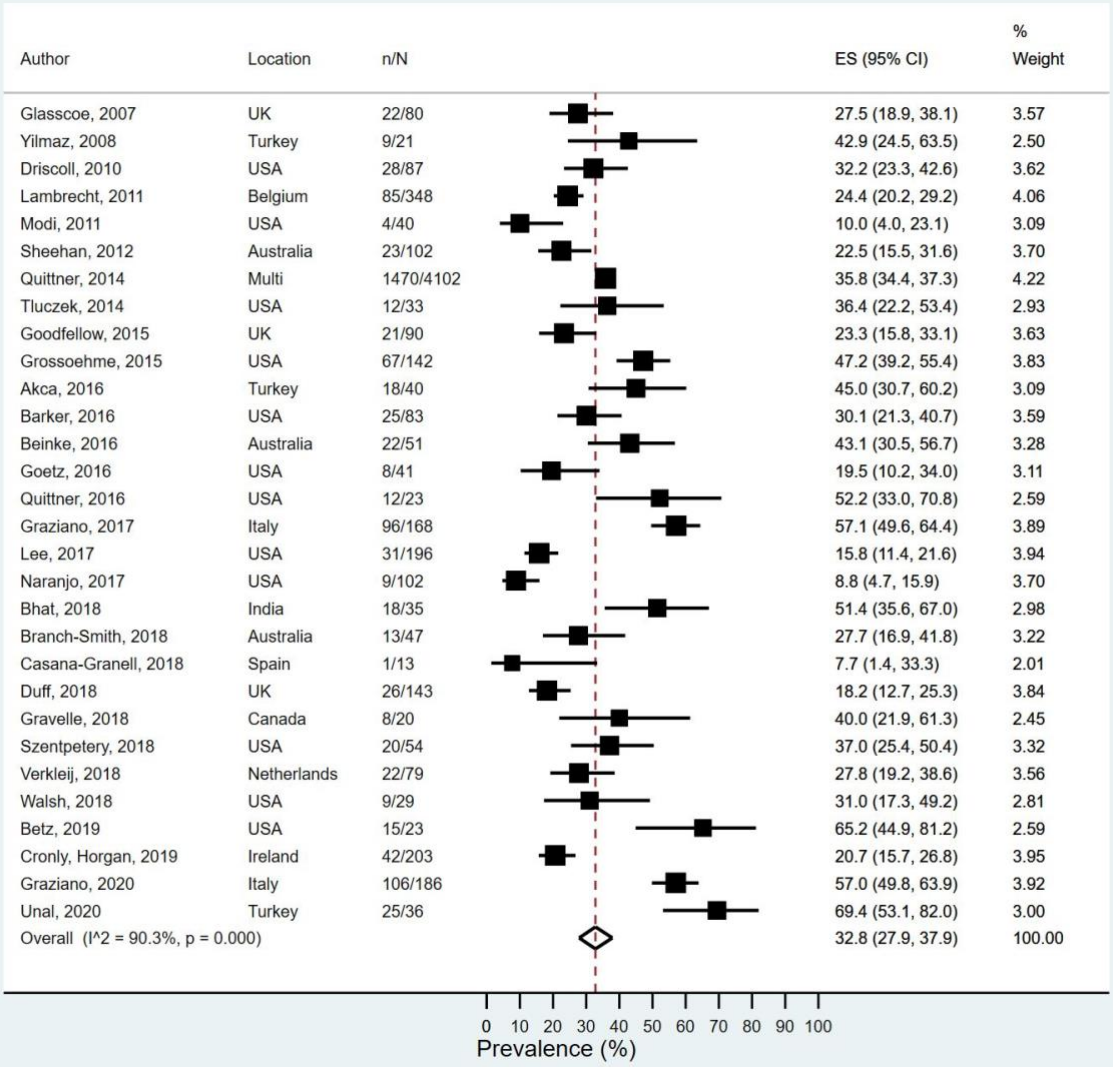

Figure S15. Forest plot of pooled prevalence of depression in caregivers of people with cystic fibrosis

Abbreviations: CI: Confidence Interval; ES: Effect Size.

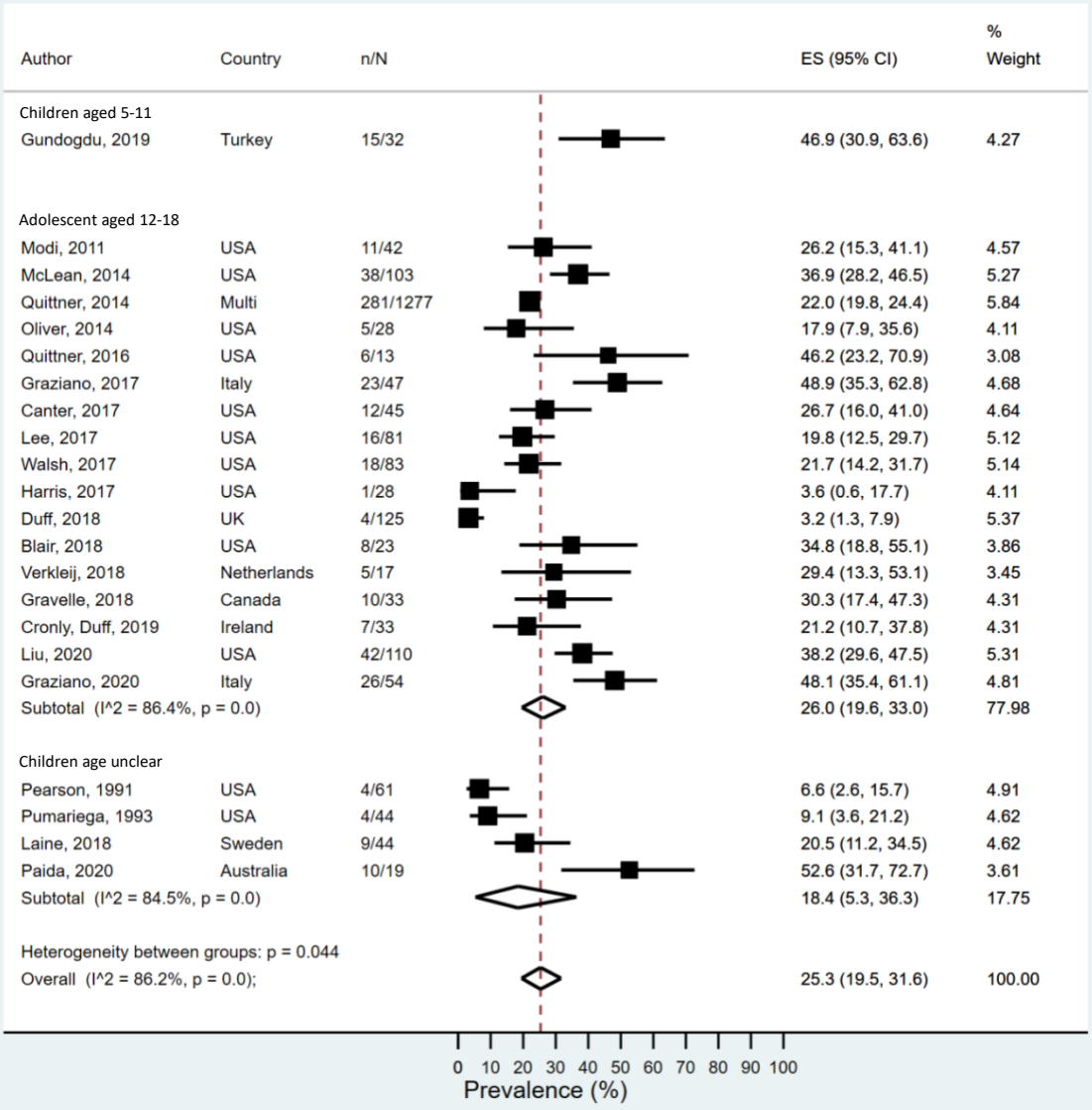

Figure S16. Forest plot of pooled prevalence of anxiety in children and adolescents with cystic fibrosis by age group

Abbreviations: CI: Confidence Interval; ES: Effect Size. Children age unclear refers to studies where participant age is not exclusive to either 5-11 or 12-18 years of age.

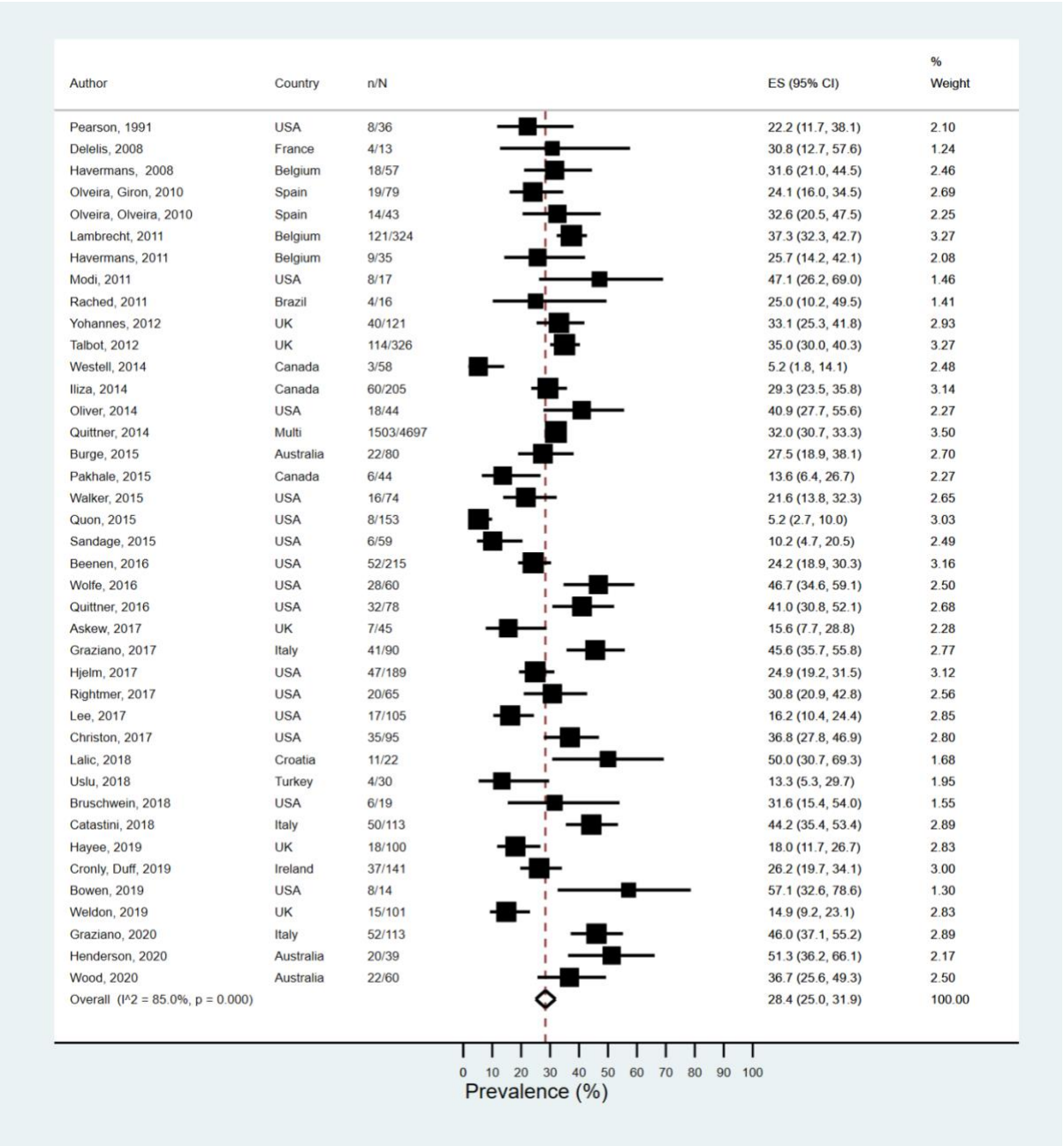

Figure S17. Forest plot of pooled prevalence of anxiety in adults with cystic fibrosis

Abbreviations: CI: Confidence Interval; ES: Effect Size.

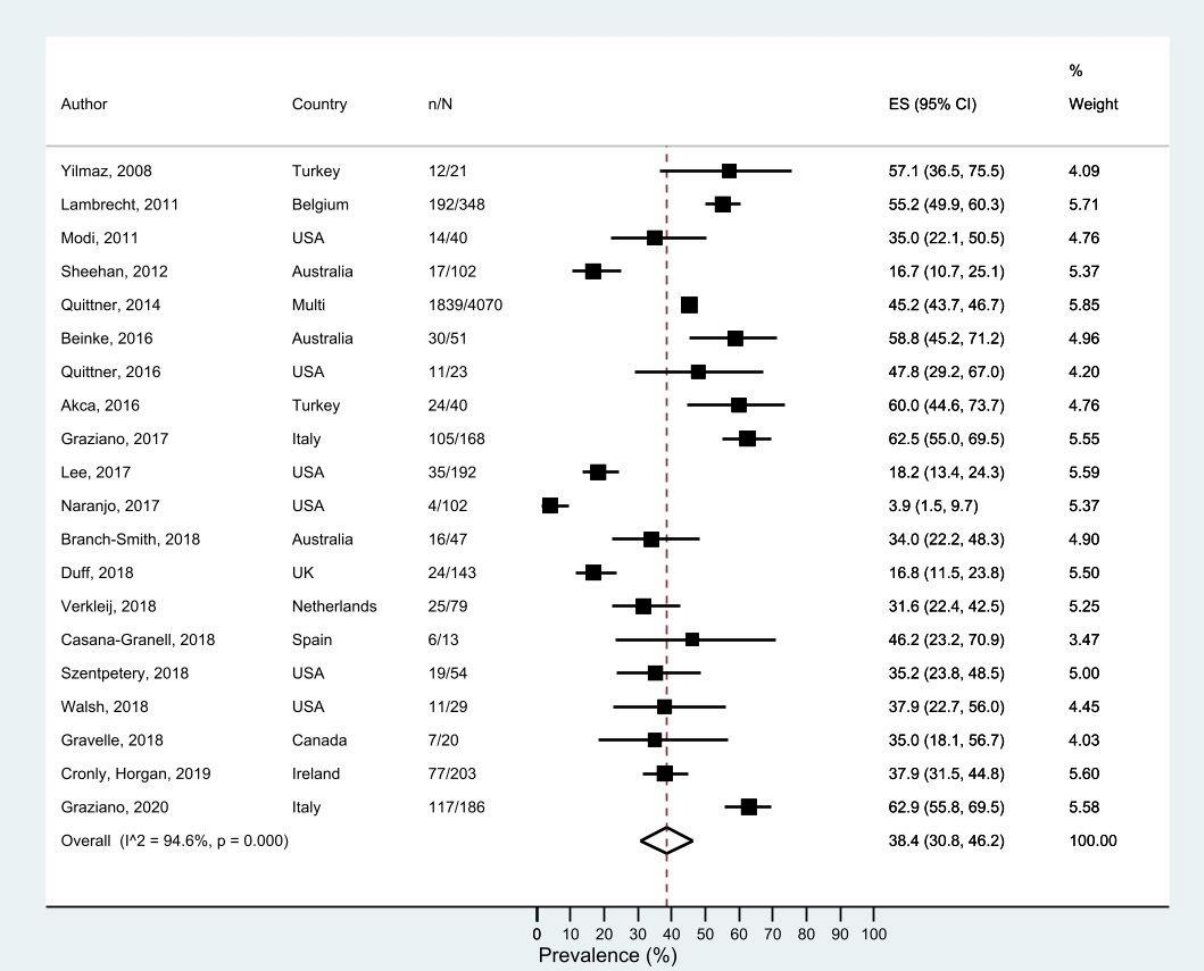

Figure S18. Forest plot of pooled prevalence of anxiety in caregivers of people with cystic fibrosis

CI: Confidence Interval; ES: Effect Size

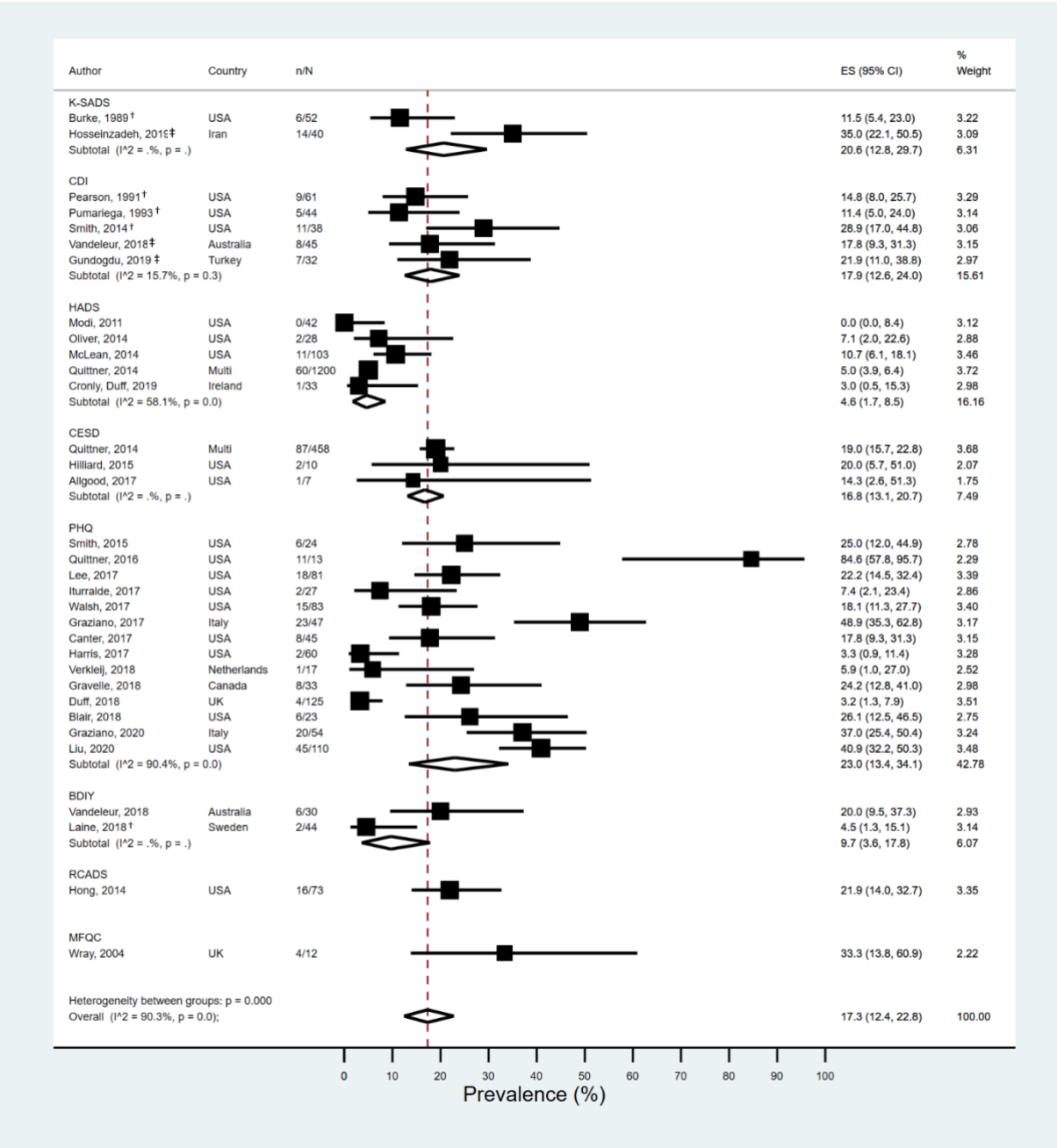

Figure S19. Forest plot of depression prevalence in children and adolescents with cystic fibrosis by psychometric tool

Abbreviations; BDIY: Beck Depression Inventory for Youth; CESD: Center for Epidemiologic Studies Depression Scale; CDI: Children's Depression Inventory; CI: Confidence Interval; ES: Effect Size; HADS: Hospital Anxiety and Depression Scale; K-SADS: Kiddie Schedule for Affective Disorders and Schizophrenia; MFQC: Mood and Feelings Questionnaire for Children; PHQ: Patient Health Questionnaire; RCADS: Revised Children's Anxiety and Depression Survey.
† Data from children with an unclear age range (where participant age is not exclusive to either 5-11 or 12-18 years of age). ‡Data from children aged 5-11 years of age.

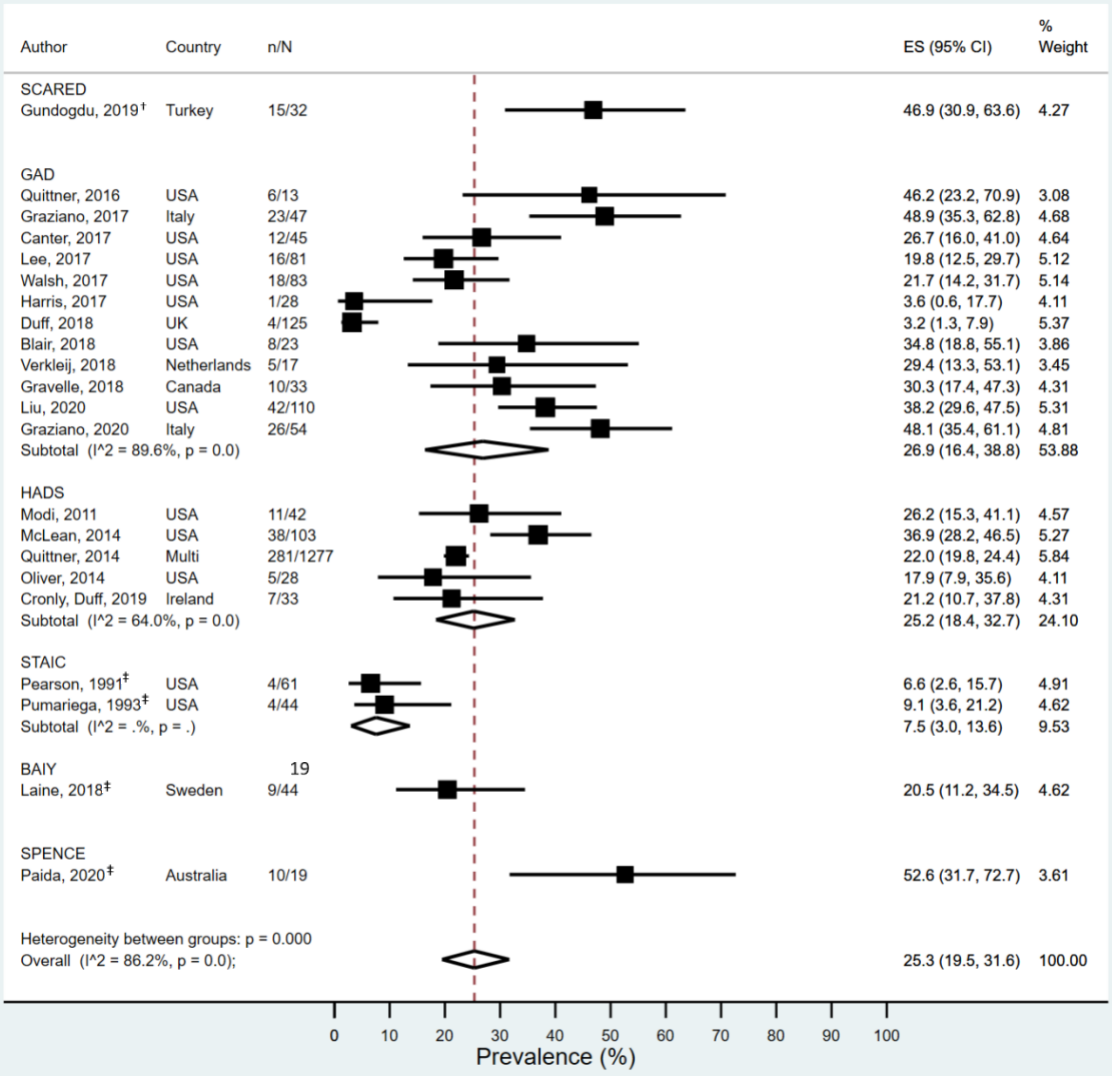

Figure S20. Forest plot of anxiety in children and adolescents with cystic fibrosis by psychometric tool

Abbreviations: BAIY: Beck Anxiety Inventory for Youth; CI: Confidence Interval; ES: Effect Size; ES: Effect Size; GAD: Generalized Anxiety Disorder 7-item measure; HADS: Hospital Anxiety and Depression Scale; SCARED: Screen for Child Anxiety Related Disorders; SPENCE: Spence Children's Anxiety Scale; STAIC: State-Trait Anxiety Inventory for Children.

†Data from children aged 5-11 years of age. ‡ Data from children with an unclear age range (where participant age is not exclusive to either 5-11 or 12-18 years of age).

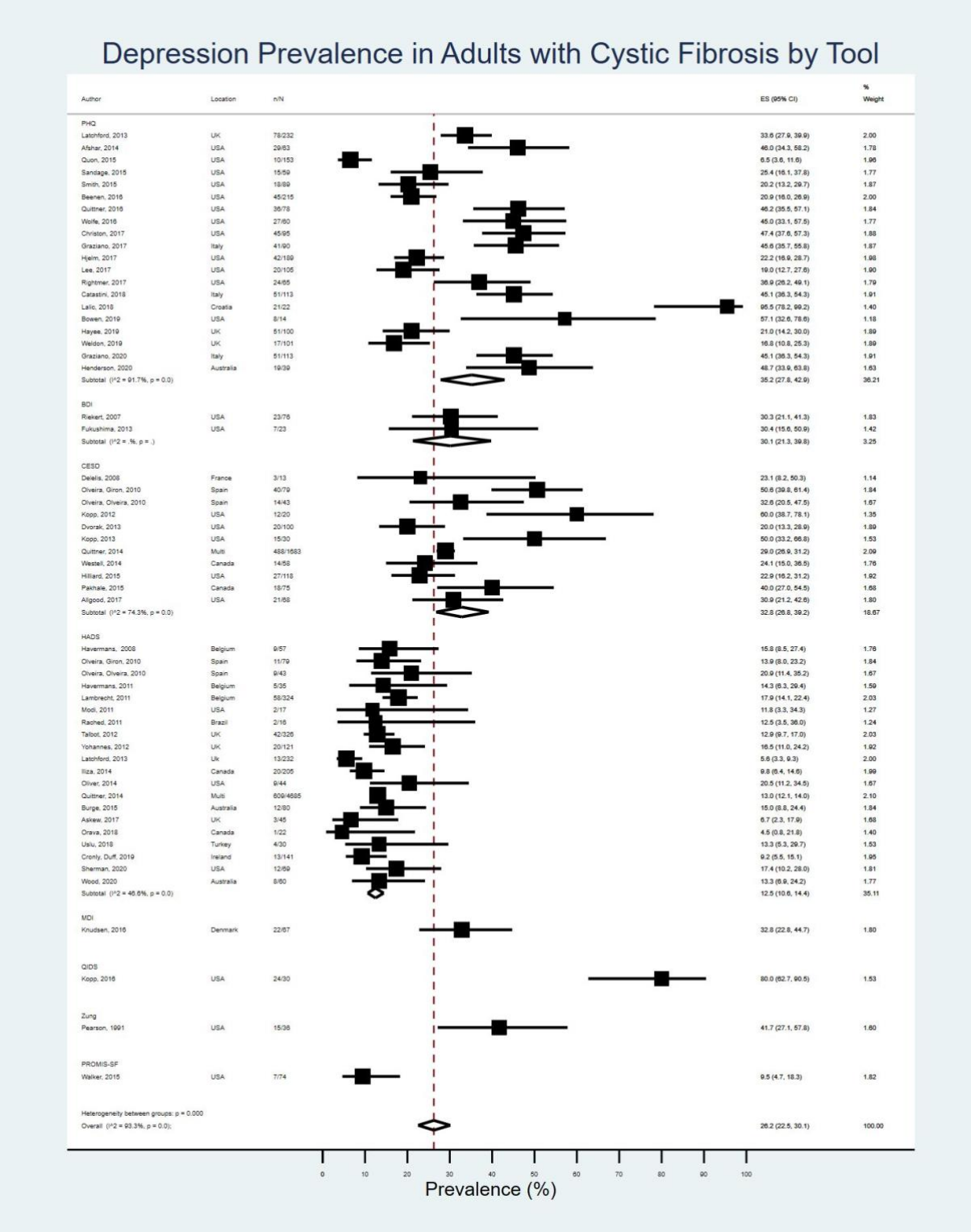

Figure S21. Forest plot of depression prevalence in adults with cystic fibrosis by psychometric tool

Abbreviations; BDI: Beck Depression Inventory; CESD: Center for Epidemiologic Studies Depression Scale; CI: Confidence Interval; ES: Effect Size; HADS: Hospital Anxiety and Depression Scale; MDI: Major Depression Inventory; PROMIS: PROMIS Depression Short Form; PHQ: Patient Health Questionnaire; QIDS: Quick Inventory of Depressive Symptomatology; Zung: Zung Self-Rating Depression Scale.

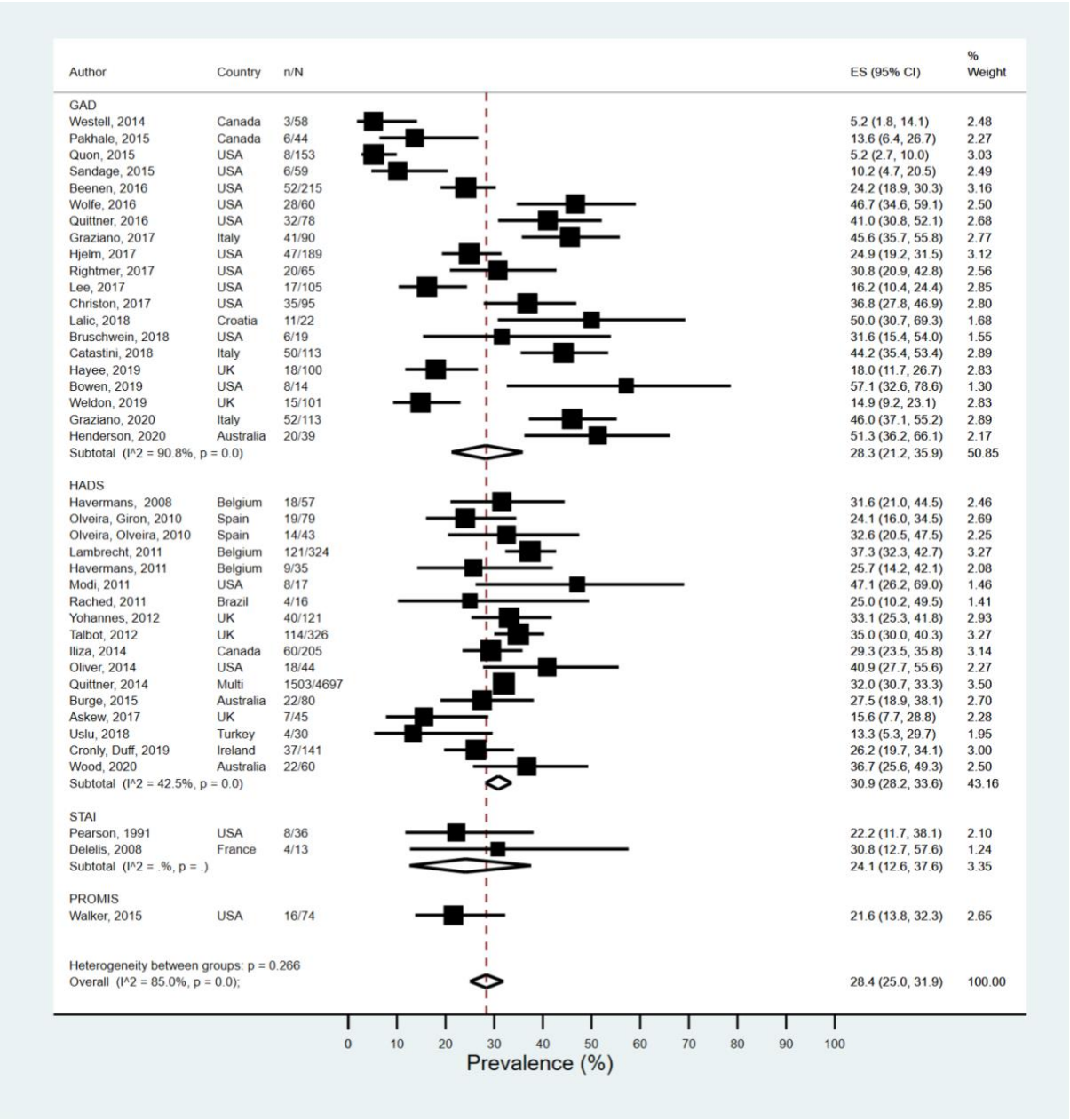

Figure S22. Forest plot of anxiety prevalence in adults with cystic fibrosis by psychometric tool

Abbreviations: CI: Confidence Interval; ES: Effect Size; GAD: Generalized Anxiety Disorder 7-item measure; HADS: Hospital Anxiety and Depression Scale; PROMIS-SF: PROMIS Anxiety Short Form; STAI: State-Trait Anxiety Inventory.

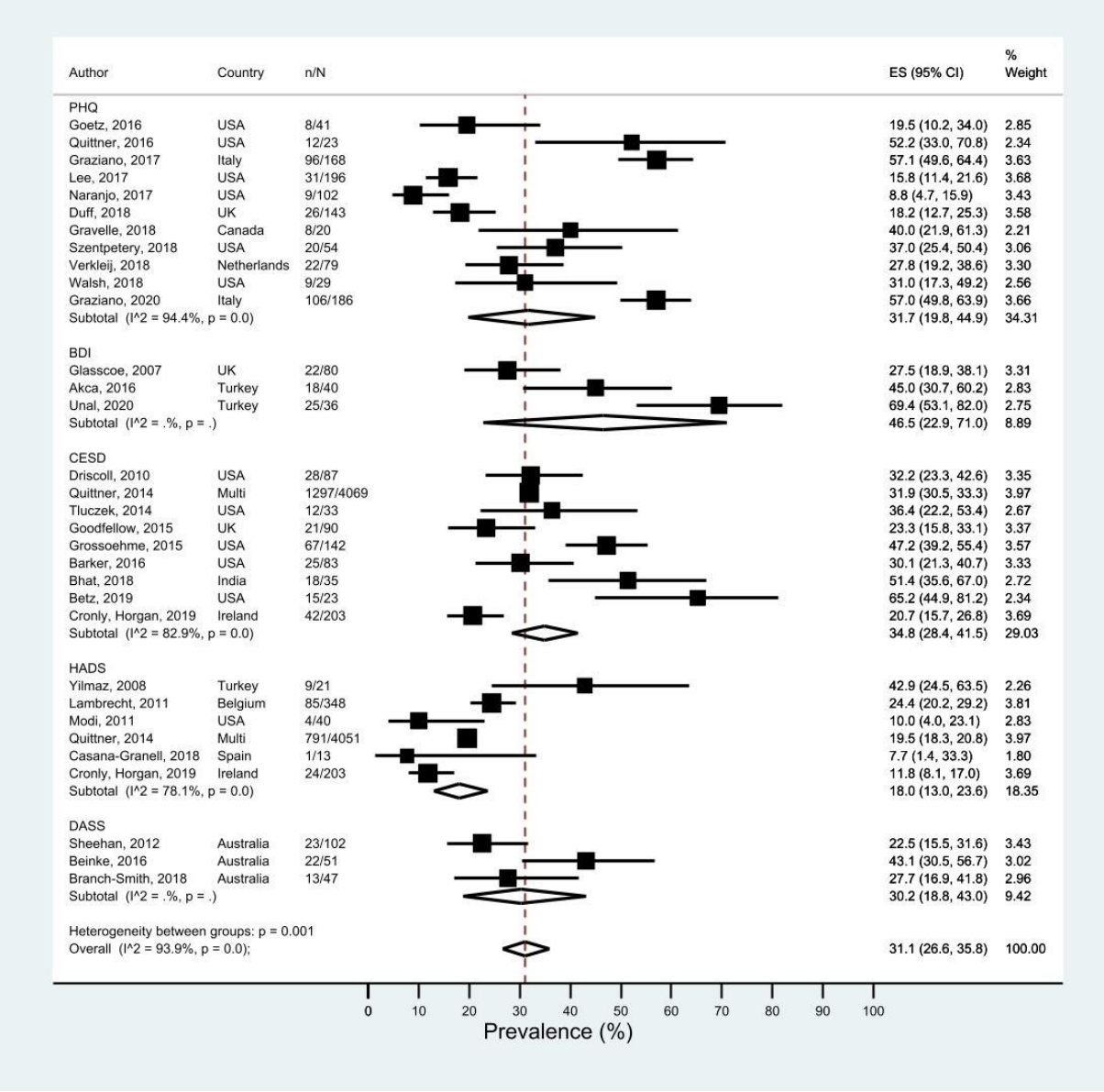

Figure S23. Forest plot of depression prevalence in caregivers of people with cystic fibrosis by psychometric tool

Abbreviations; BDI: Beck Depression Inventory; CESD: Center for Epidemiologic Studies Depression Scale; CI: Confidence Interval; DASS: Depression Anxiety Stress Scale; ES: Effect Size; HADS: Hospital Anxiety and Depression Scale; PHQ: Patient Health Questionnaire.

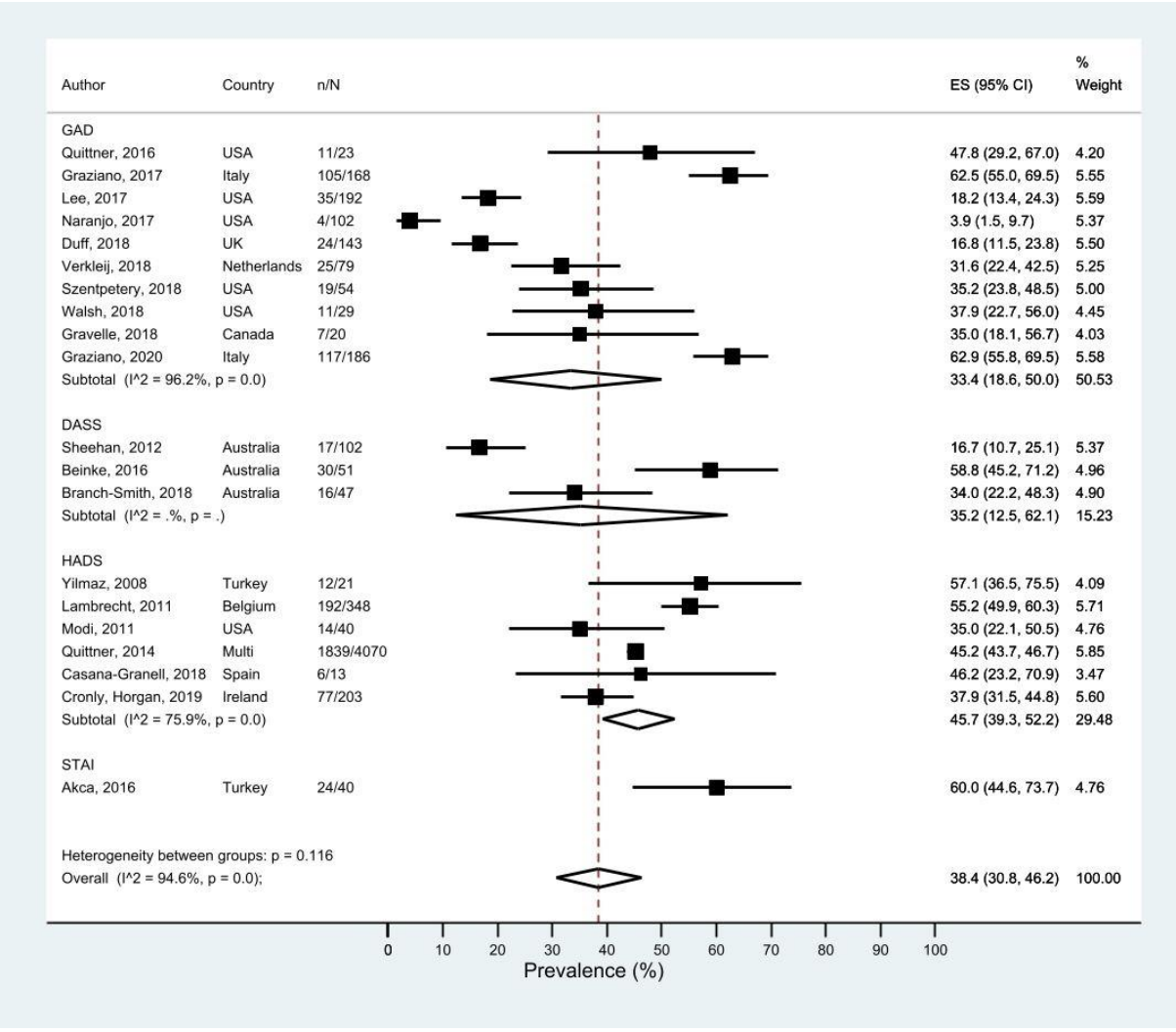

Figure S24. Forest plot of anxiety in caregivers of people with cystic fibrosis by psychometric tool

Abbreviations: CI: Confidence Interval; DASS: Depression Anxiety Stress Scale; ES: Effect Size; GAD: Generalized Anxiety Disorder 7-item measure; HADS: Hospital Anxiety and Depression Scale; STAI: State-Trait Anxiety Inventory.

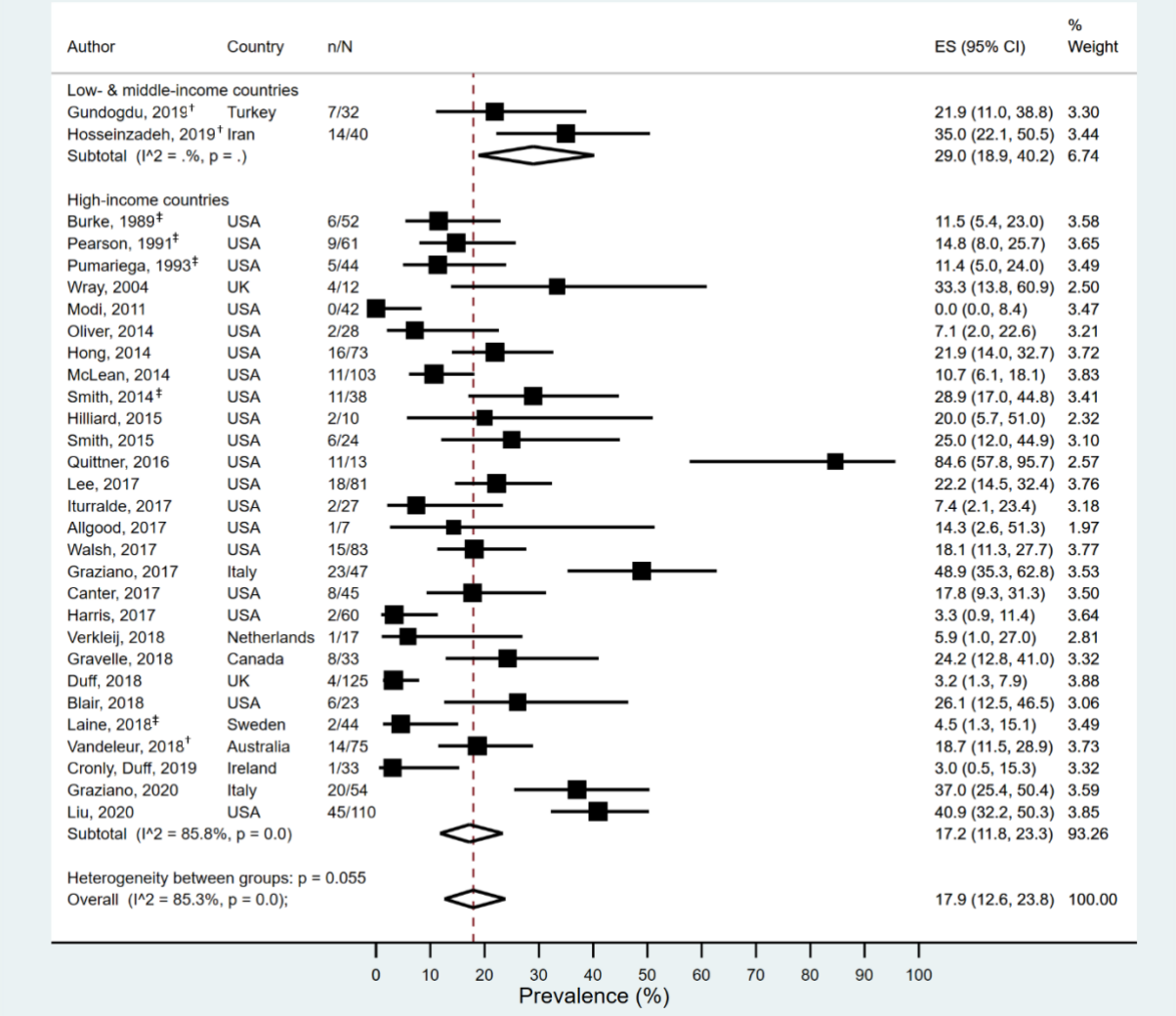

Figure S25. Forest plot of depression in children and adolescents with cystic fibrosis by study location

Abbreviations: CI: Confidence Interval; ES: Effect Size.
†Data from children aged 5-11 years of age. ‡ Data from participants with an unclear age range (where participant age is not exclusive to either 5-11 or 12-18 years of age).

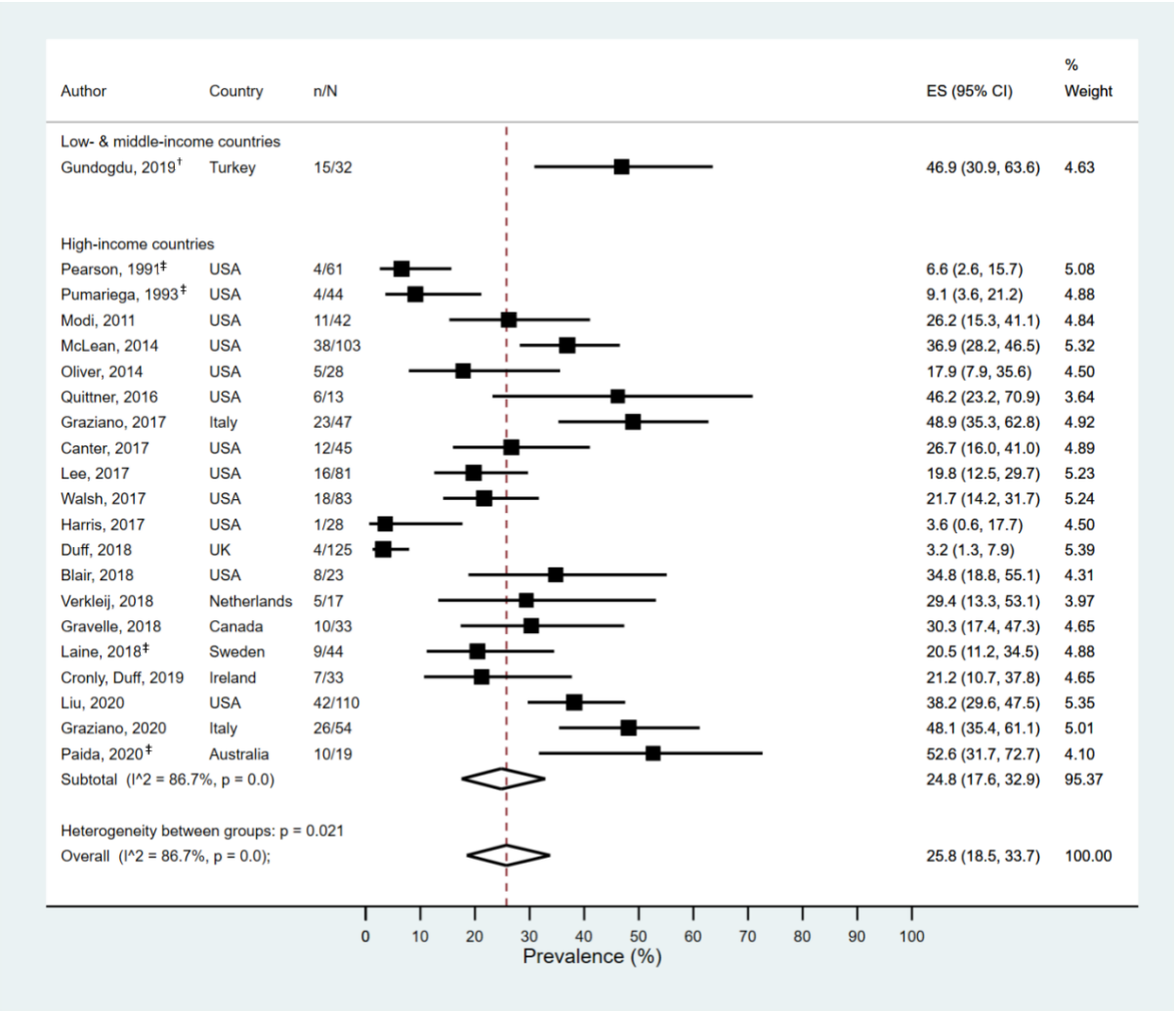

Figure S26. Forest plot of anxiety in children and adolescents with cystic fibrosis by study location

Abbreviations: CI: Confidence Interval; ES: Effect Size.
†Data from children aged 5-11 years of age. ‡ Data from participants with an unclear age range (where participant age is not exclusive to either 5-11 or 12-18 years of age).

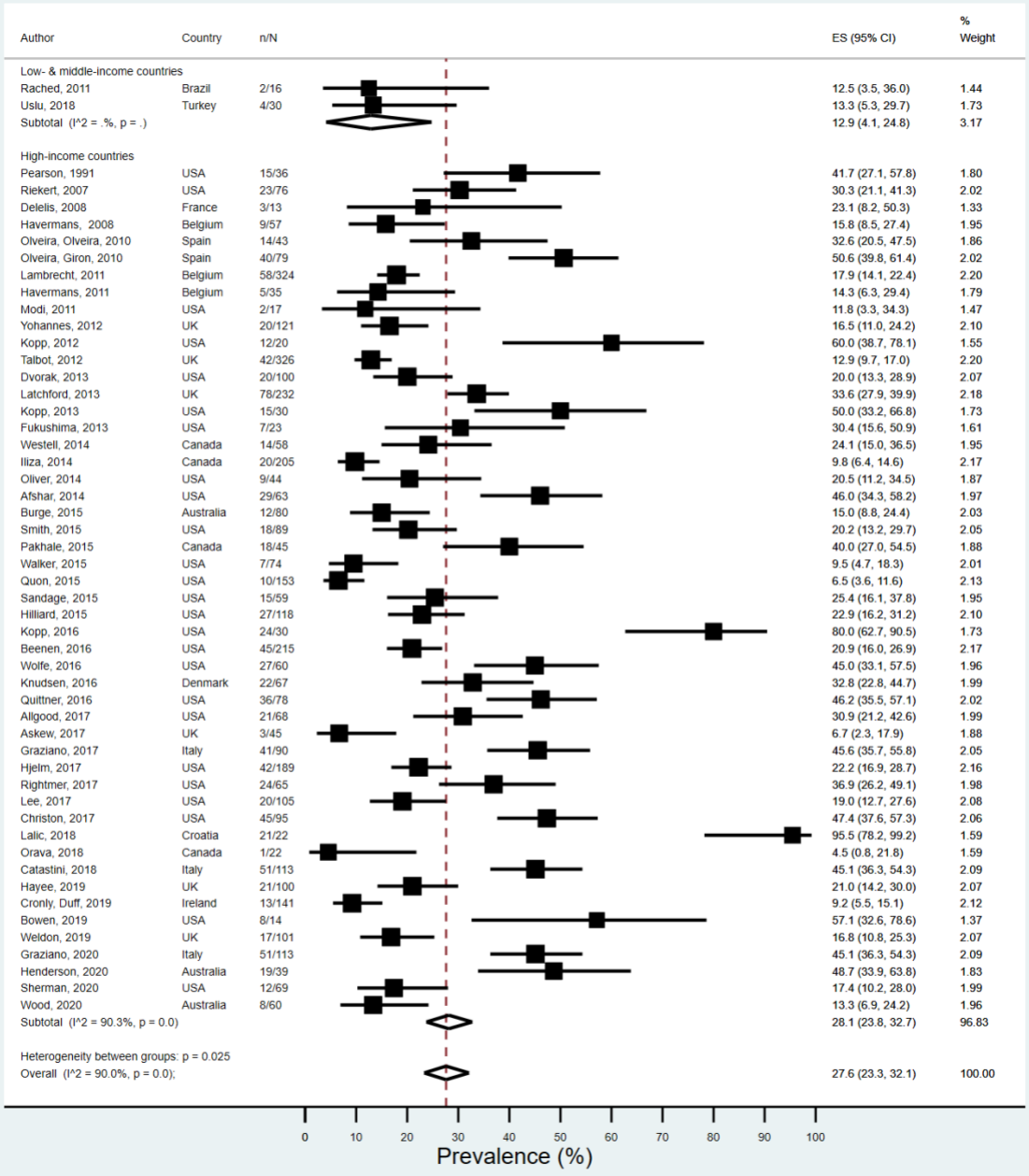

Figure S27. Forest plot of depression in adults with cystic fibrosis by study location

Abbreviations: CI: Confidence Interval; ES: Effect Size.

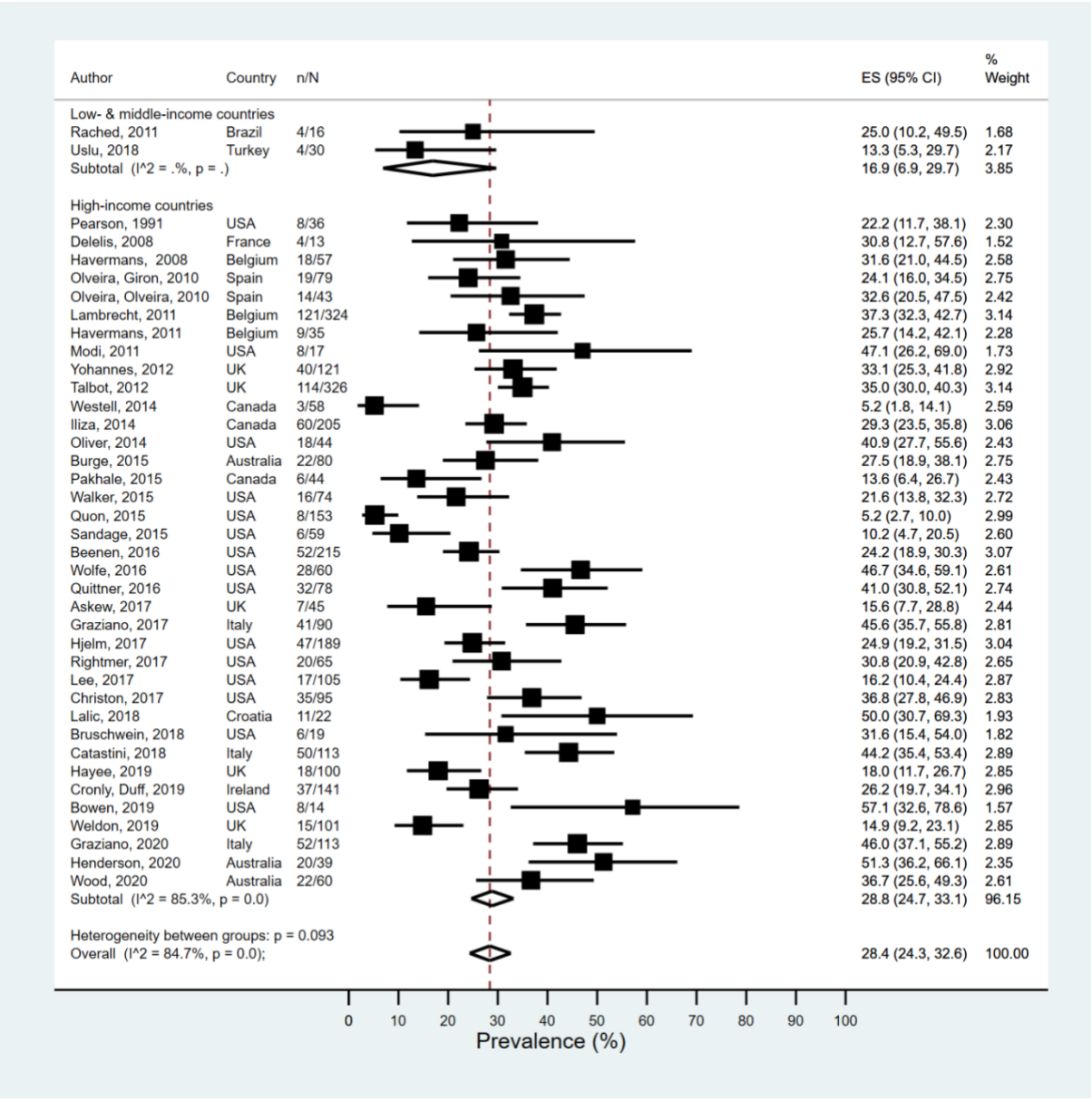

Figure S28. Forest plot of anxiety in adults with cystic fibrosis by study location

Abbreviations: CI: Confidence Interval; ES: Effect Size.

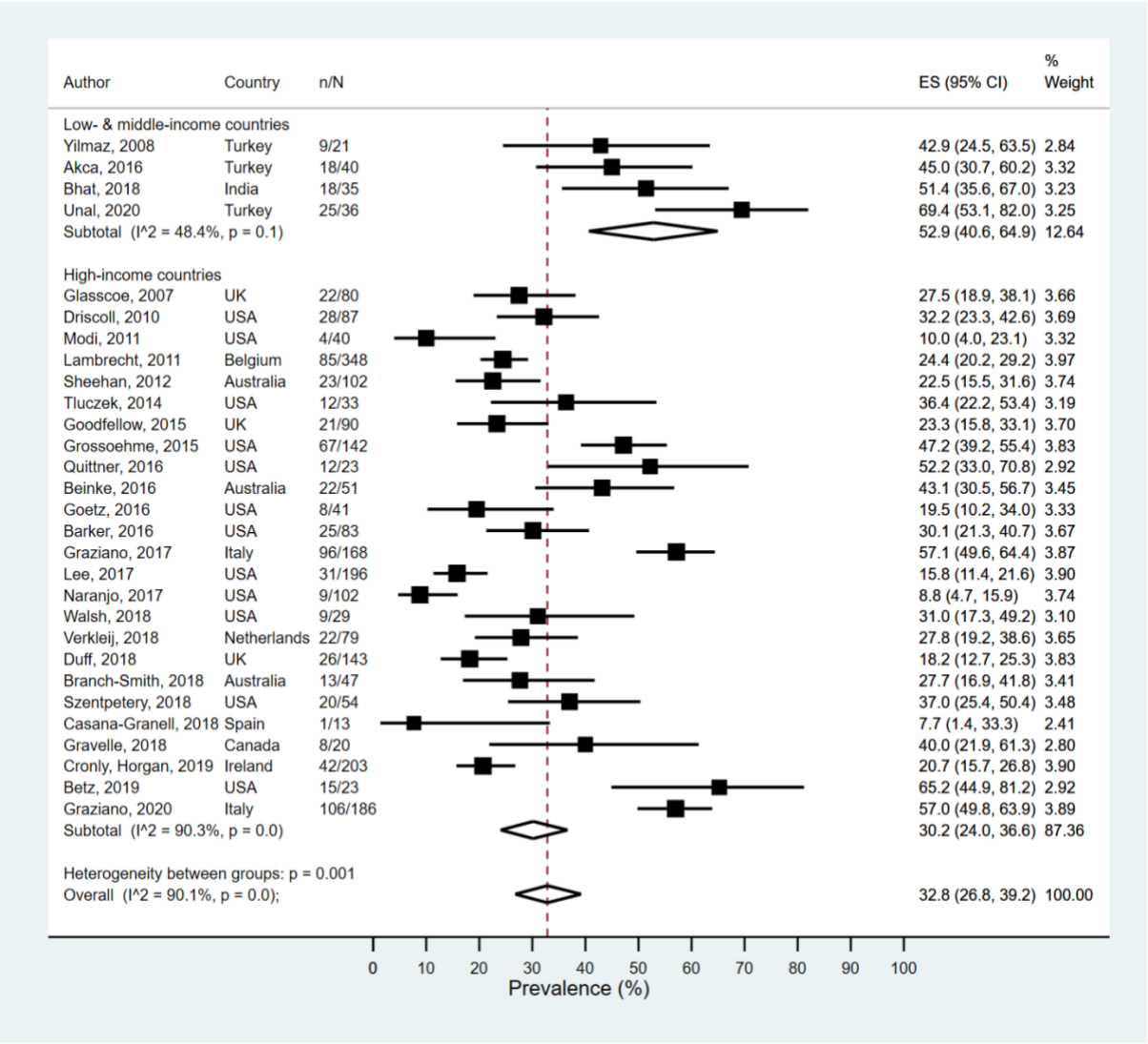

Figure S29. Forest plot of depression in caregivers of people with cystic fibrosis by study location

Abbreviations: CI: Confidence Interval; ES: Effect Size.

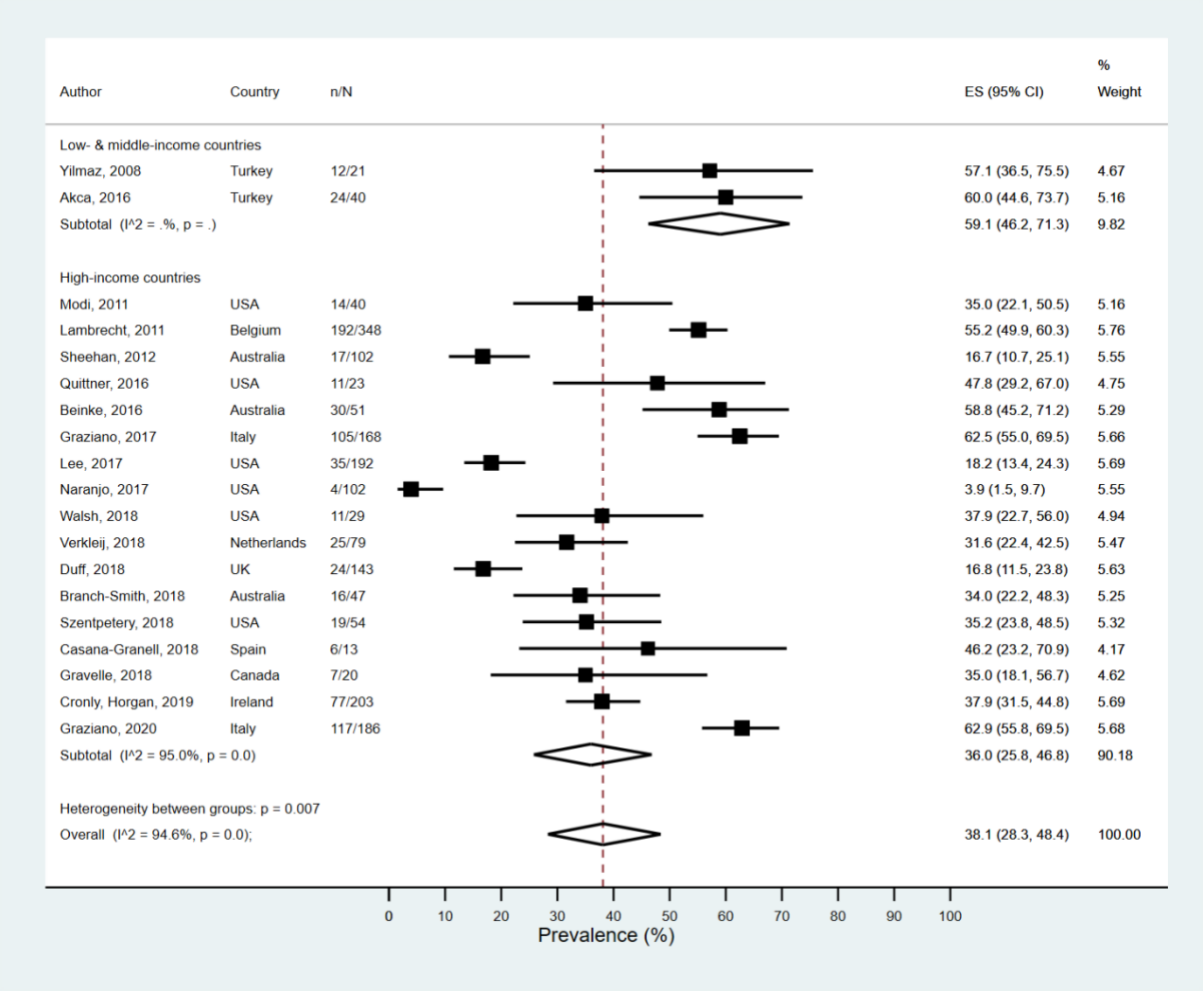

Figure S30. Forest plot of anxiety in caregivers of people with cystic fibrosis by study location

Abbreviations: CI: Confidence Interval; ES: Effect Size.

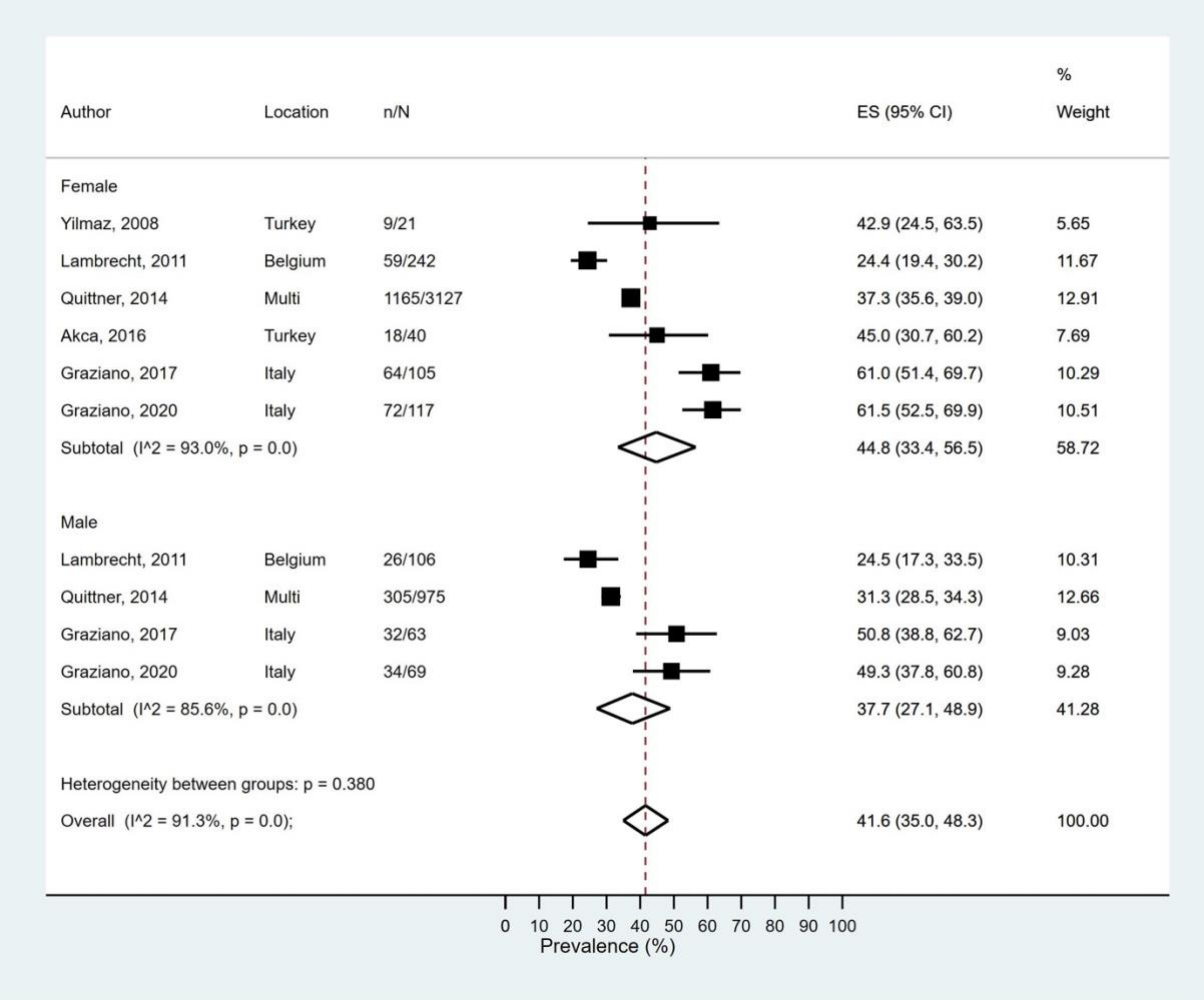

Figure S31. Forest plot of depression in caregivers of people with cystic fibrosis by caregiver sex

Abbreviations: CI: Confidence Interval; ES: Effect Size.

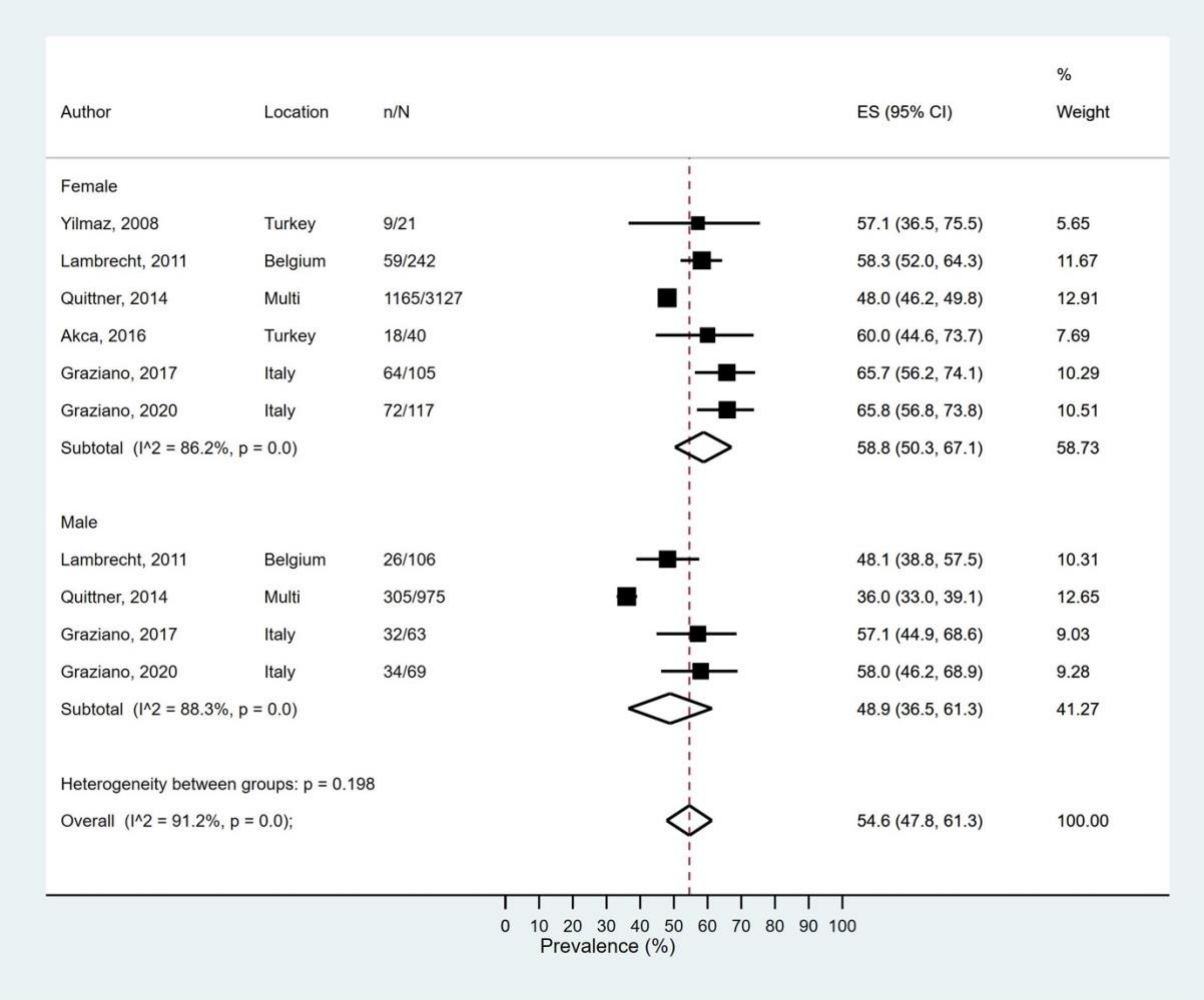

Figure S32. Forest plot of anxiety in caregivers of people with cystic fibrosis by caregiver sex

Abbreviations: CI: Confidence Interval; ES: Effect Size.

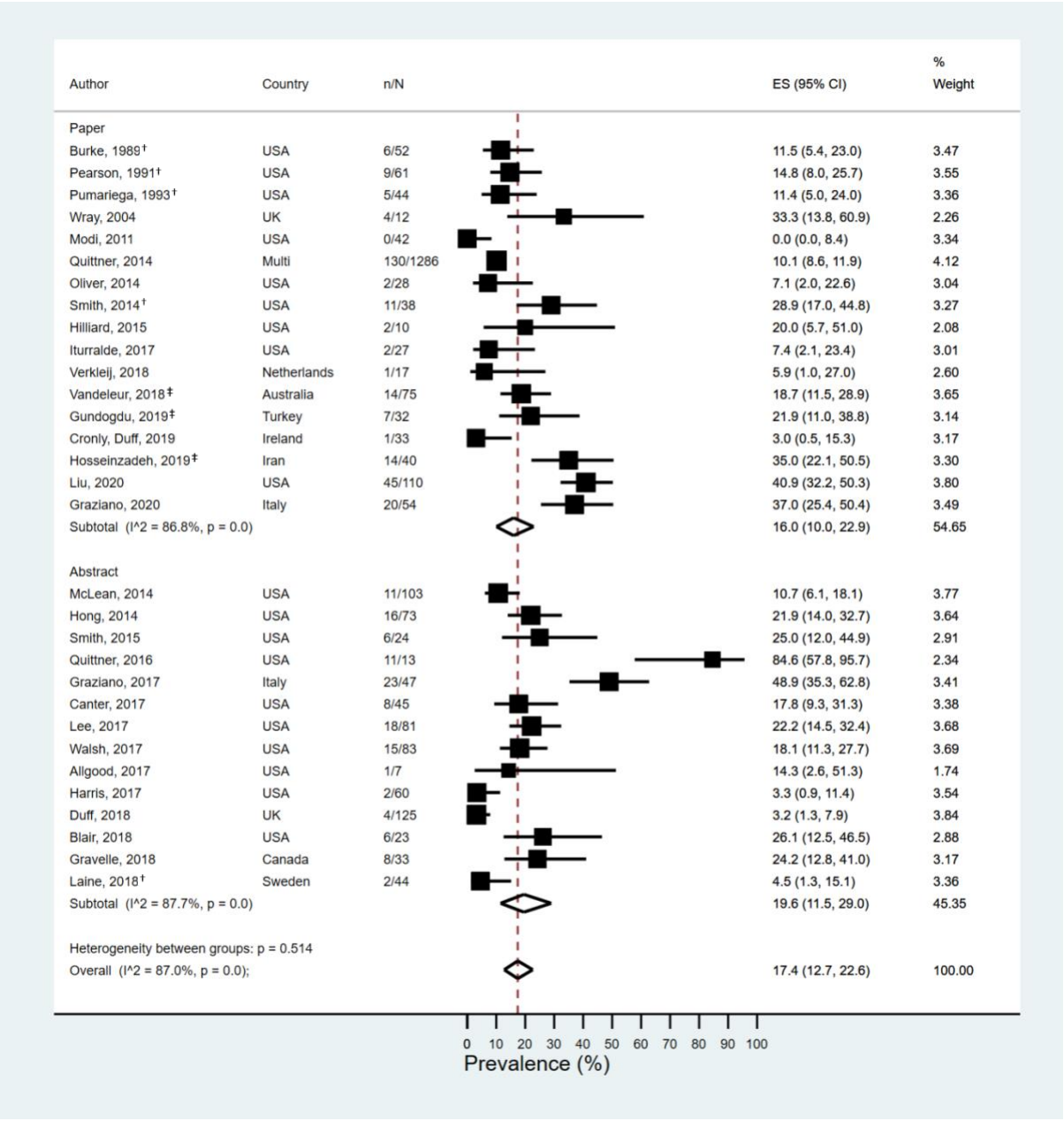

Figure S33. Forest plot of depression in children and adolescents with cystic fibrosis by publication type

Abbreviations: CI: Confidence Interval; ES: Effect Size; Paper: full-text publication. †Data from participants with an unclear age range (where participant age is not exclusive to either 5-11 or 12-18 years of age). ‡Data from children aged 5-11 years of age.

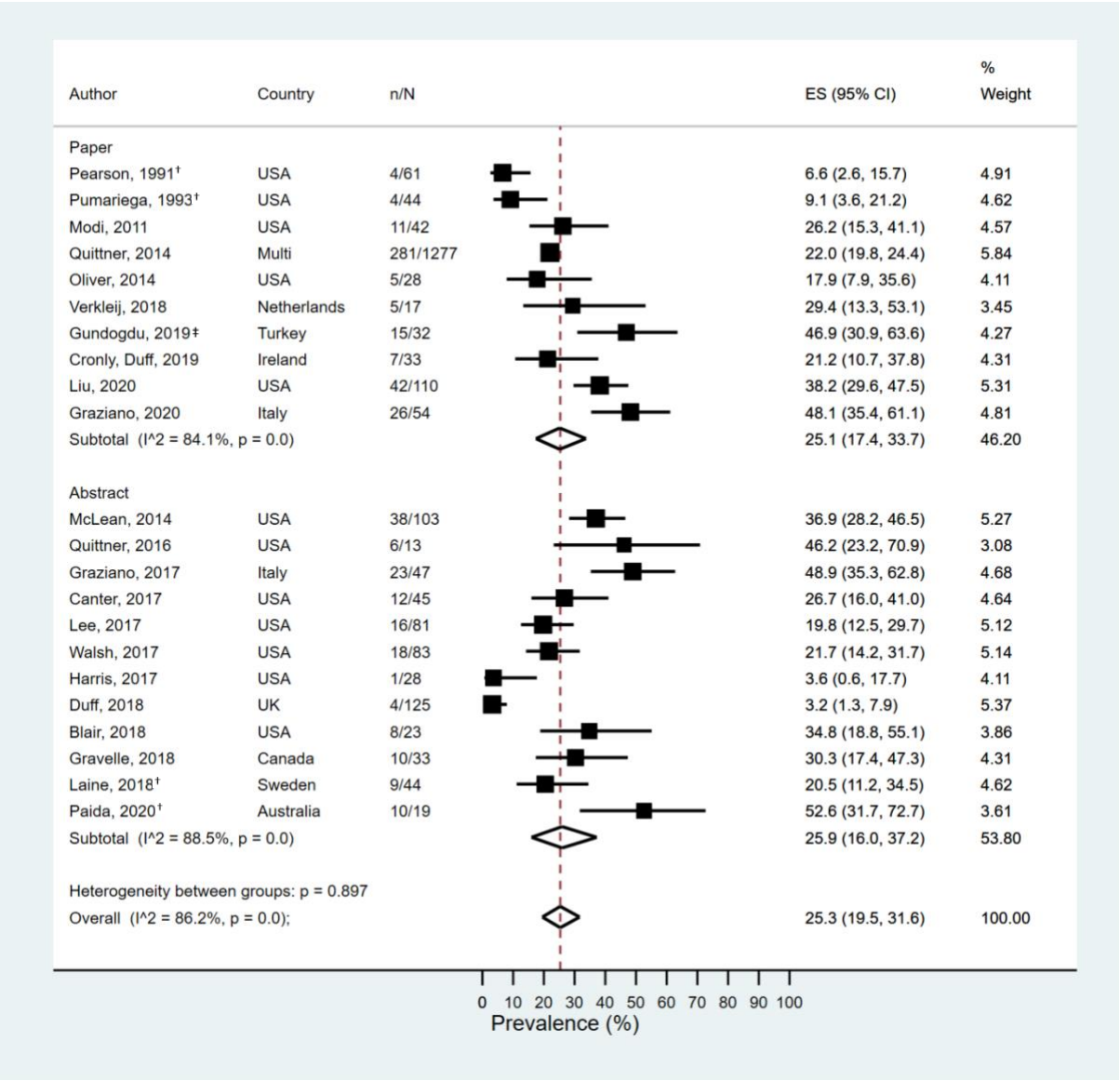

Figure S34. Forest plot of anxiety in children and adolescents with cystic fibrosis by publication type

Abbreviations: CI: Confidence Interval; ES: Effect Size; Paper: full-text publication. †Data from participants with an unclear age range (where participant age is not exclusive to either 5-11 or 12-18 years of age). ‡Data from children aged 5-11 years of age.

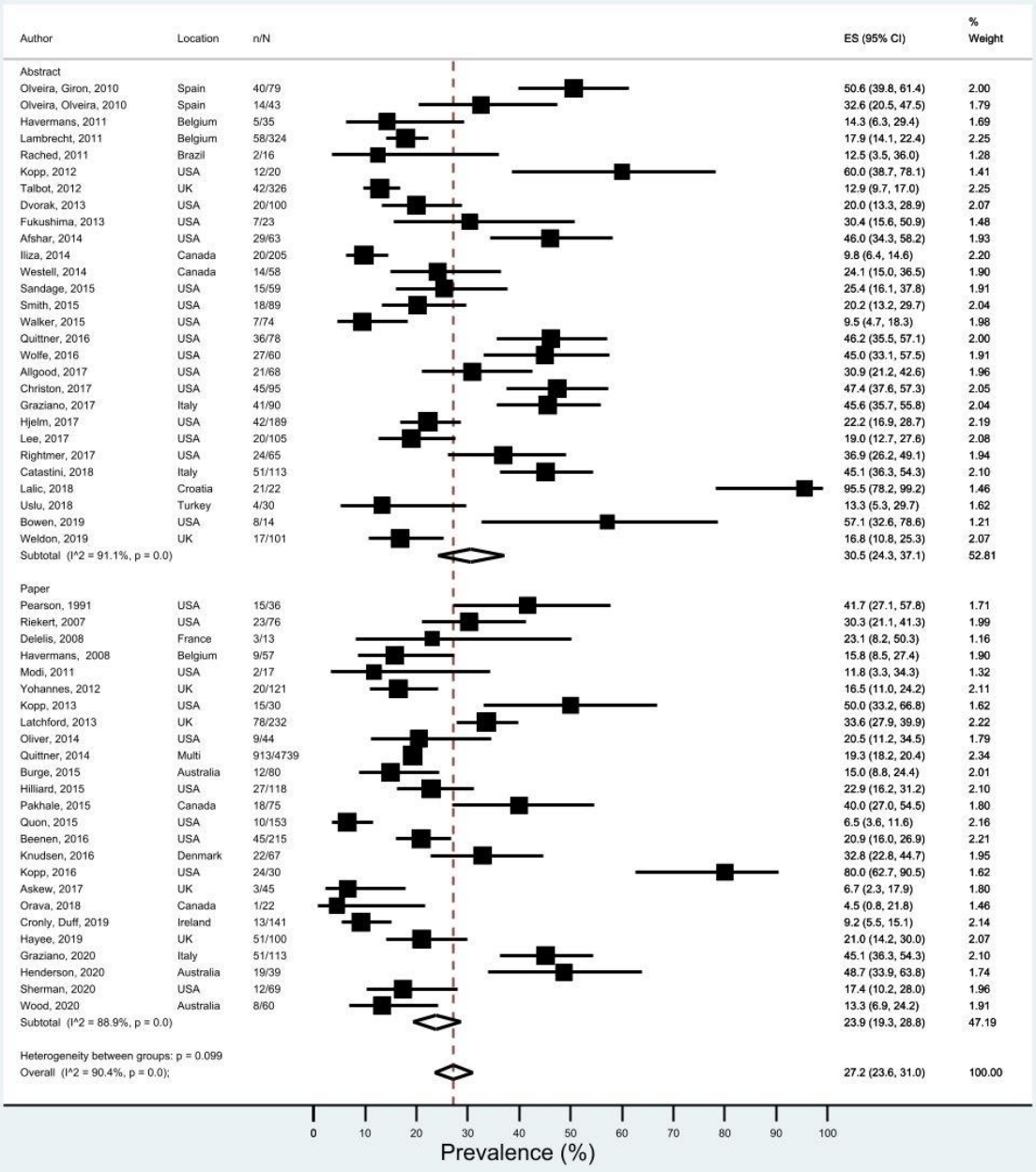

Figure S35. Forest plot of depression in adults with cystic fibrosis by publication type

Abbreviations: CI: Confidence Interval; ES: Effect Size; Paper: full-text publication.

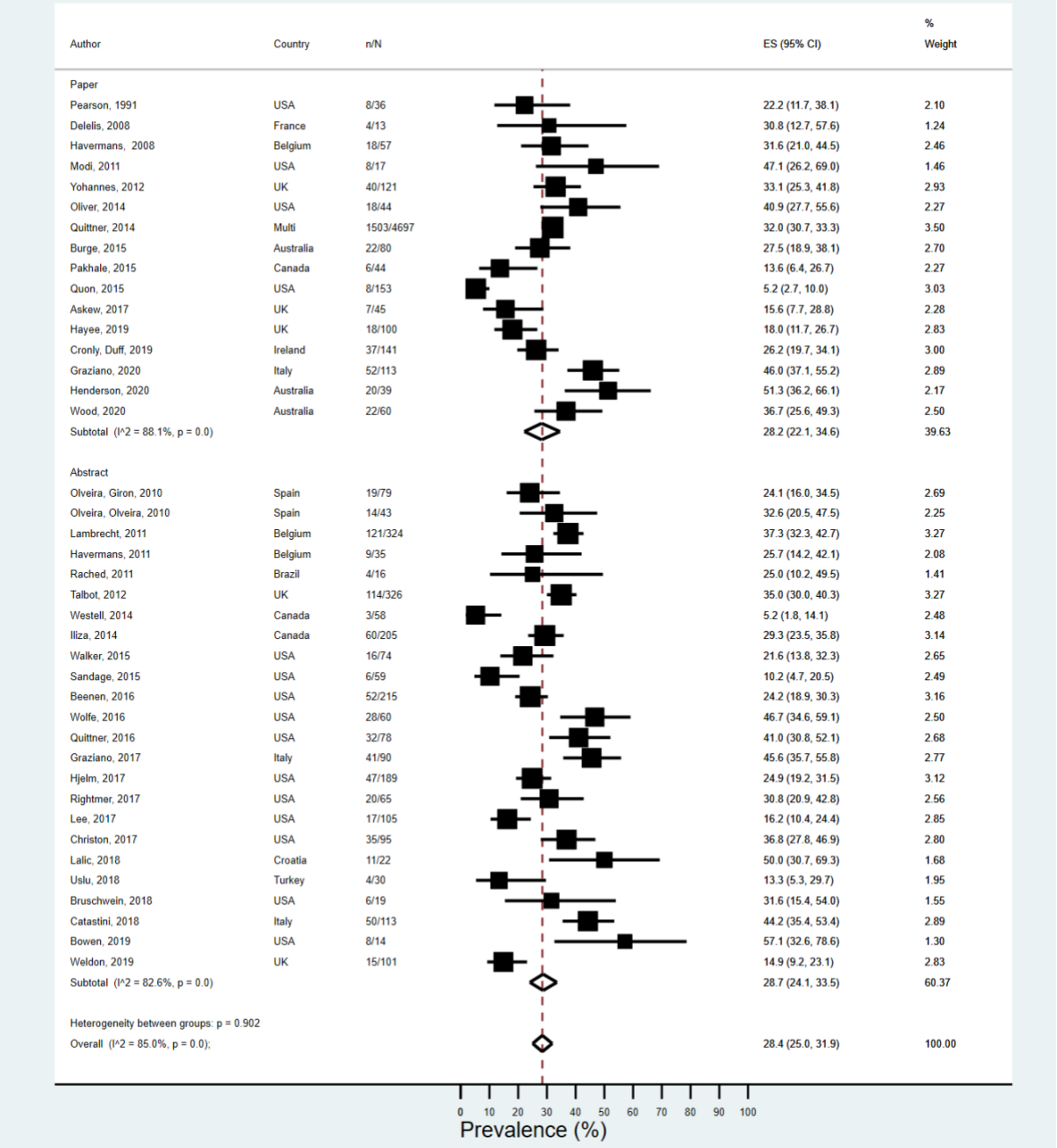

Figure S36. Forest plot of anxiety in adults with cystic fibrosis by publication type

Abbreviations: CI: Confidence Interval; ES: Effect Size; Paper: full-text publication.

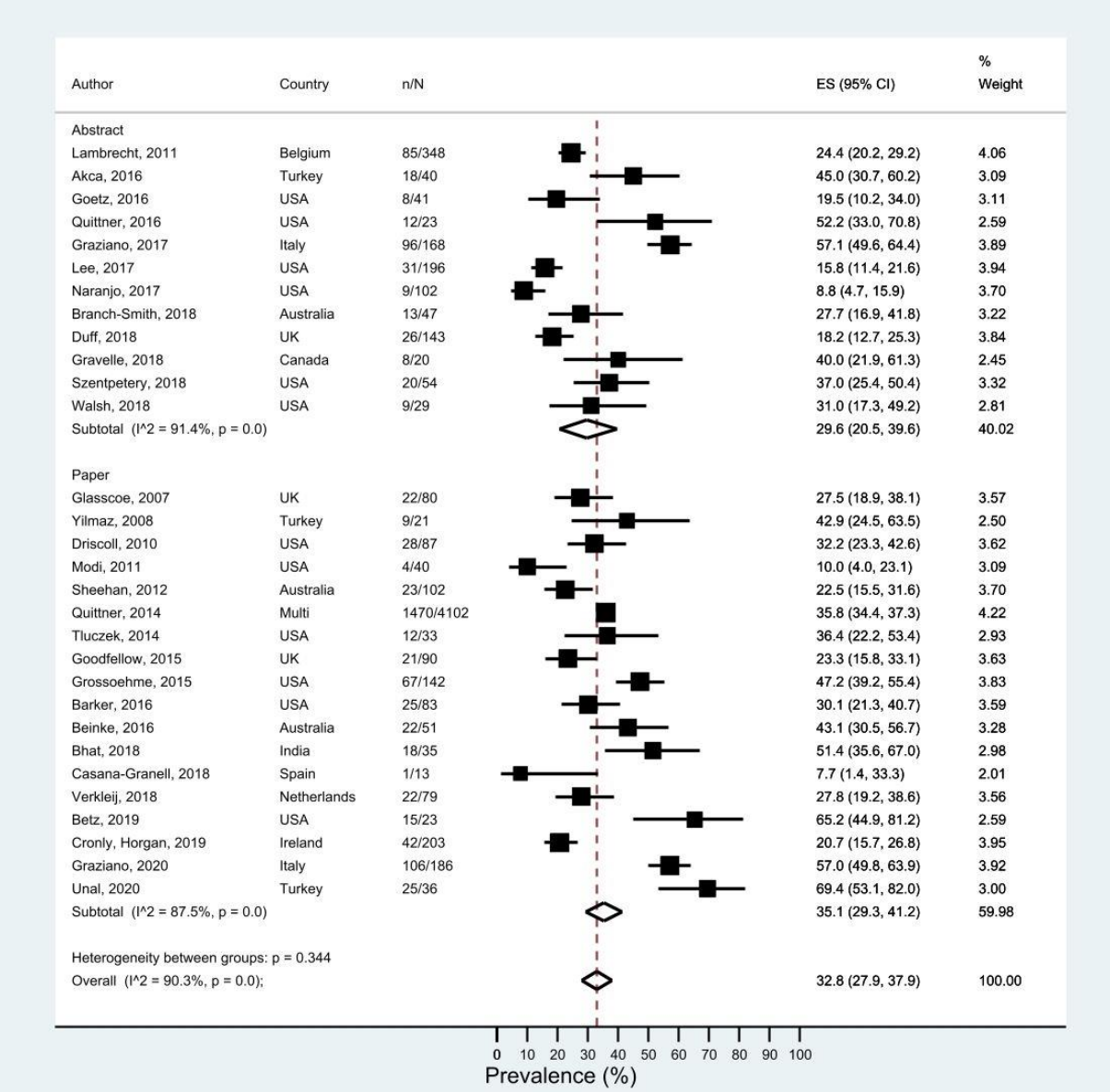

Figure S37. Forest plot of depression in caregivers of people with cystic fibrosis by publication type

Abbreviations: CI: Confidence Interval; ES: Effect Size; Paper: full-text publication.

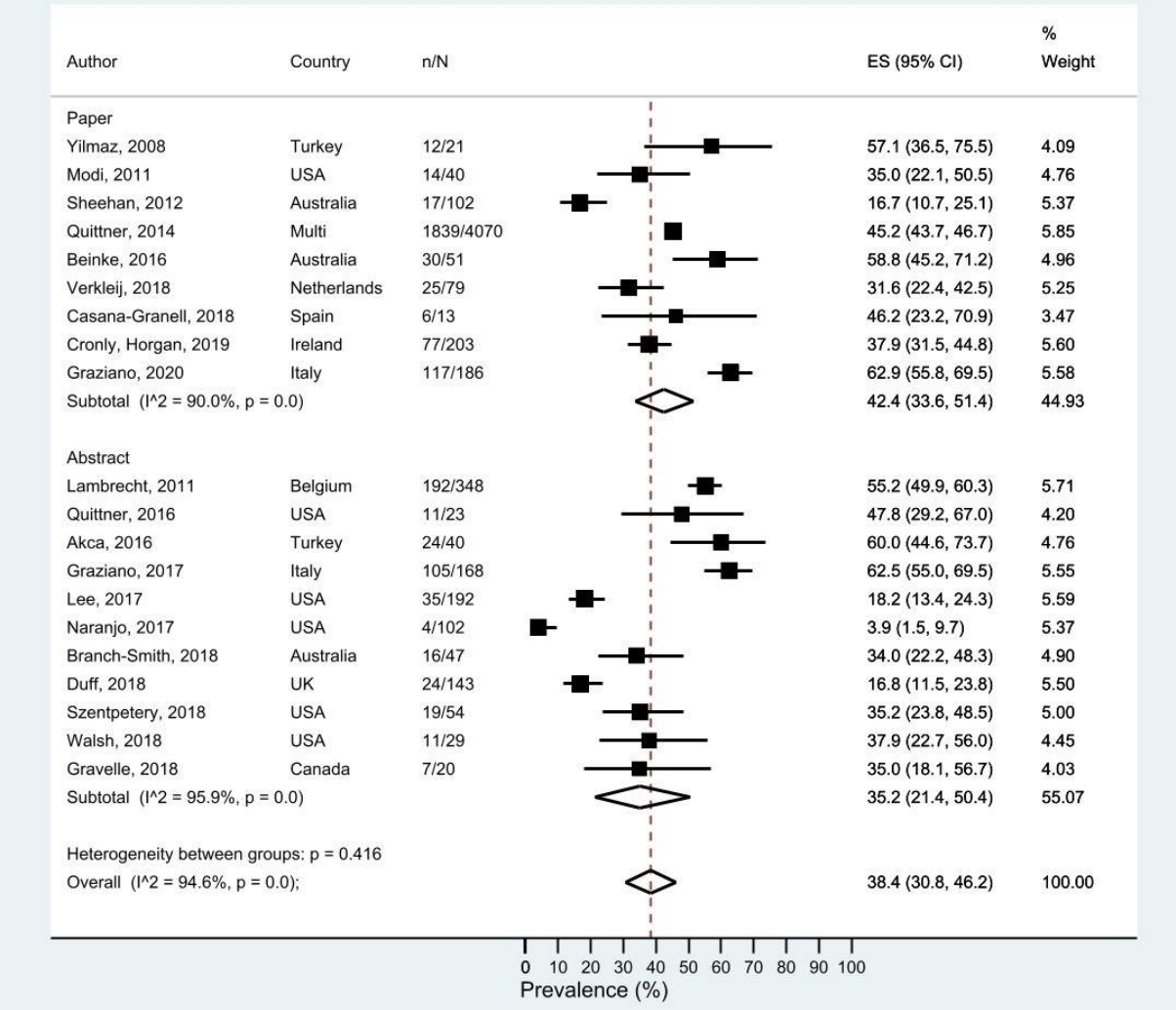

Figure S38. Forest plot of anxiety in caregivers of people with cystic fibrosis by publication type

Abbreviations: CI: Confidence Interval; ES: Effect Size; Paper: full-text publication.

Table S10. PRISMA Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Title
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Abstract
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Introduction/1-3
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Introduction/3
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Methods/3
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Methods/2
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Supplementary information
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Methods/3-4
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Methods/5-7
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Methods/5-7
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Methods/5-7
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Methods/6
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Methods/7
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Methods/7
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data	Methods/7

Section and Topic	Item #	Checklist item	Location where item is reported
		conversions.	
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Methods/7
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Methods/7
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Methods/7
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Methods/7
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Methods/7
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Methods/7
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Results/1; Figure 1; Supplementary Information
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	Table S1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Table S5-S9; Figure S1-S6
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Figure S13-S38
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Results/1-15
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Results/4-13; Figure 2-5; Figure S13-S38
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Results/3-13; Figure S19-S38
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Results/6
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Results/3; Table S5- S9;

Section and Topic	Item #	Checklist item	Location where item is reported
			Figure S1-S12
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Results; Figures S13-S38
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Discussion/1-5
	23b	Discuss any limitations of the evidence included in the review.	Discussion/6-7
	23c	Discuss any limitations of the review processes used.	Discussion/7
	23d	Discuss implications of the results for practice, policy, and future research.	Discussion1-7; Conclusion
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Methods/1
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Methods/1
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Not applicable
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Funding
Competing interests	26	Declare any competing interests of review authors.	Conflict of interest
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Available on request from author

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